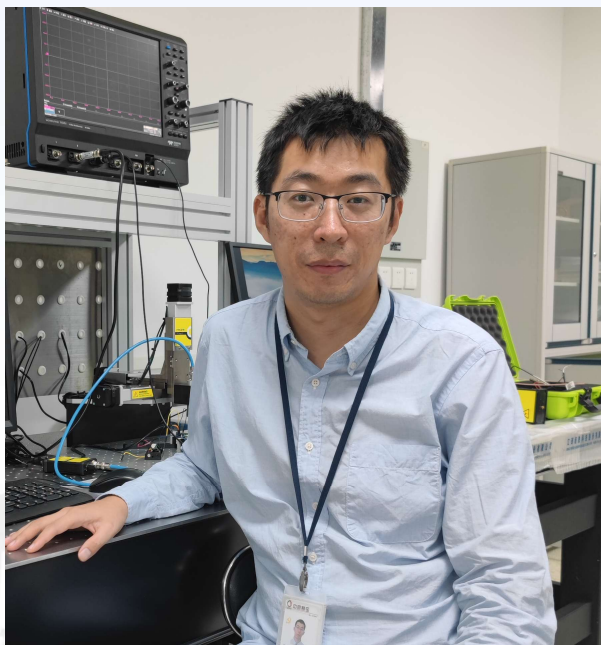




数字集成电路设计--第六章

CMOS组合逻辑门设计

➤ 个人简介



雷登云 教授

个人简介:

2022.07-今 集成电路学院, 副教授、教授。

2015.7-2022.7 工业和信息化部电子第五研究所 高级工程师

2010.9-2015.7 北京大学, 微电子学与固体电子学博士

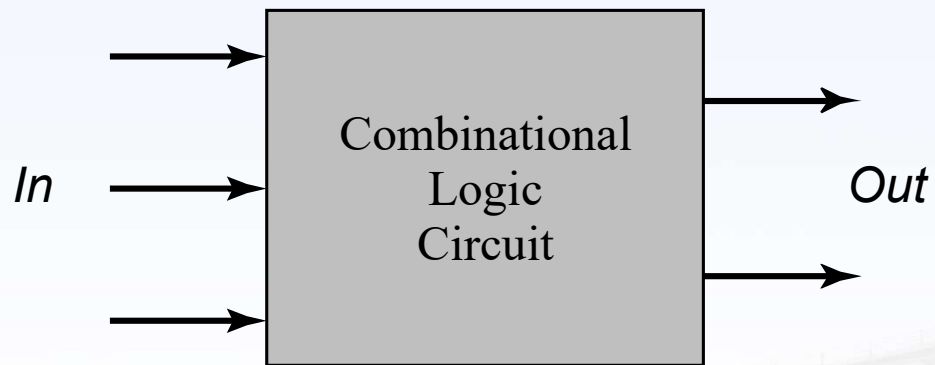
2006.9-2010.7 西安电子科技大学 集成电路设计与集成系统

研究方向: 软硬件协同设计、集成电路安全性和可靠性

办公室地址: 大学城校区理学馆710-3室

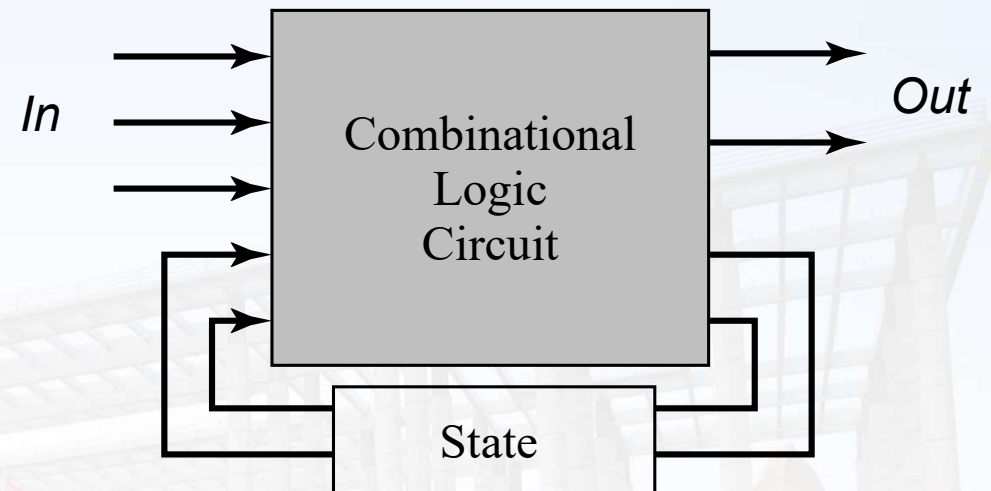
邮箱: leidy@gdut.edu.cn

组合电路和时序电路



组合电路

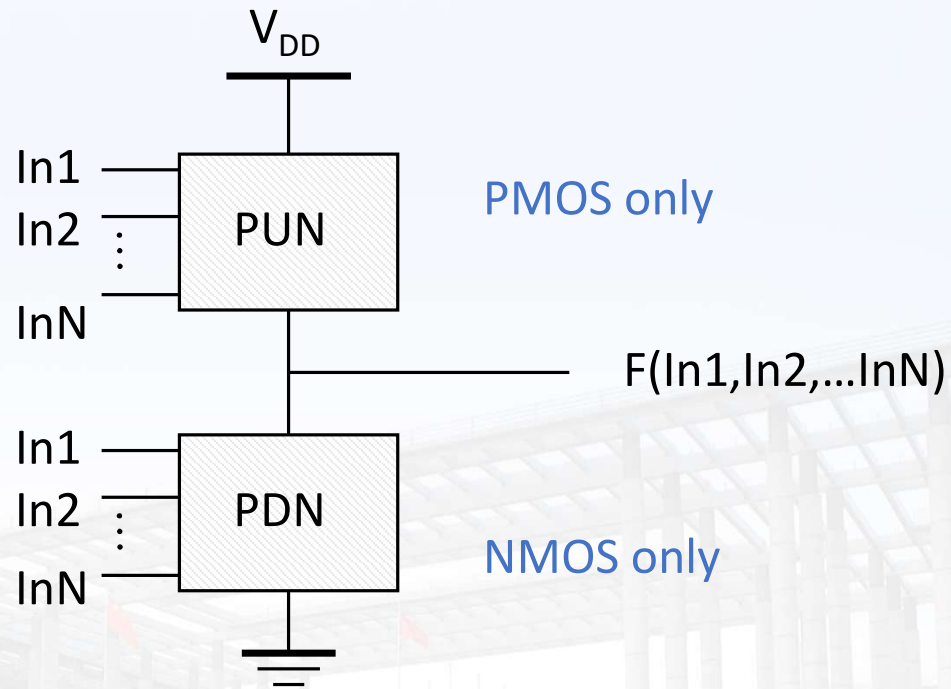
$$\text{Output} = f(\text{In})$$



时序电路

$$\text{Output} = f(\text{In}, \text{Previous In})$$

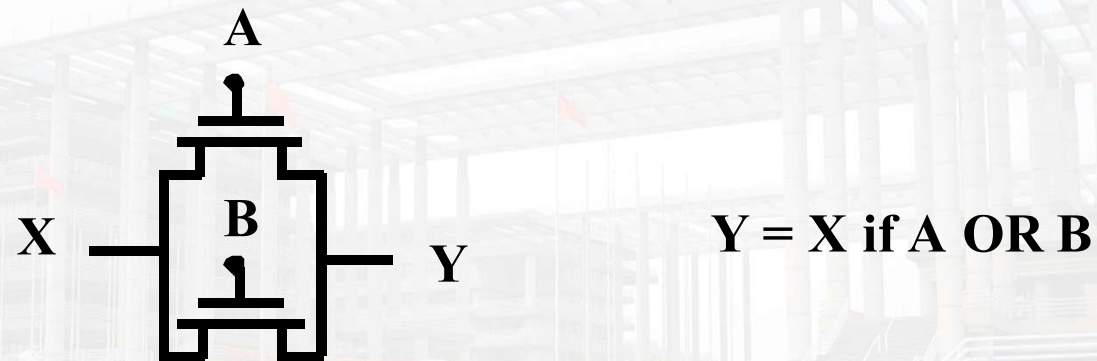
静态互补CMOS



PUN和PDN组成互补逻辑门结构

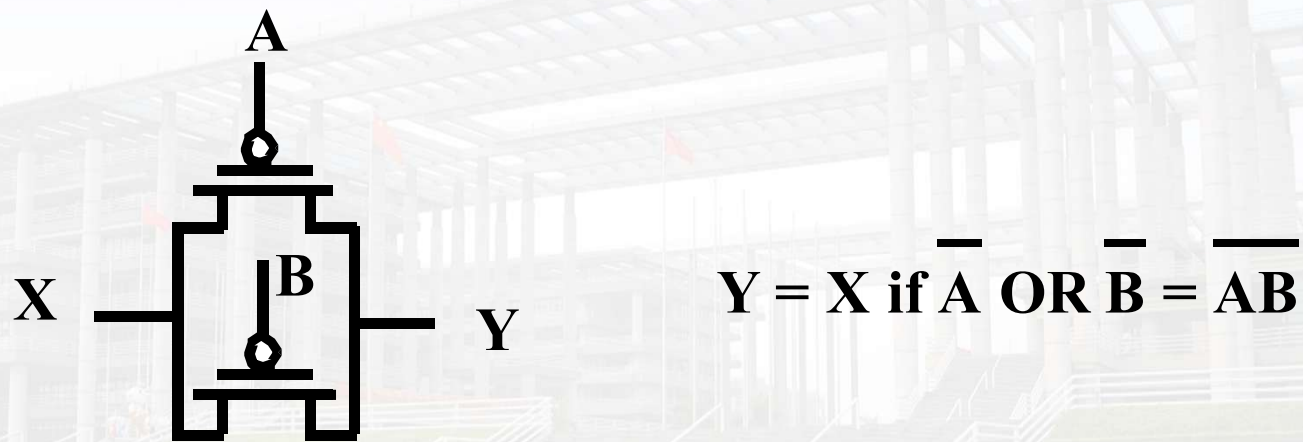
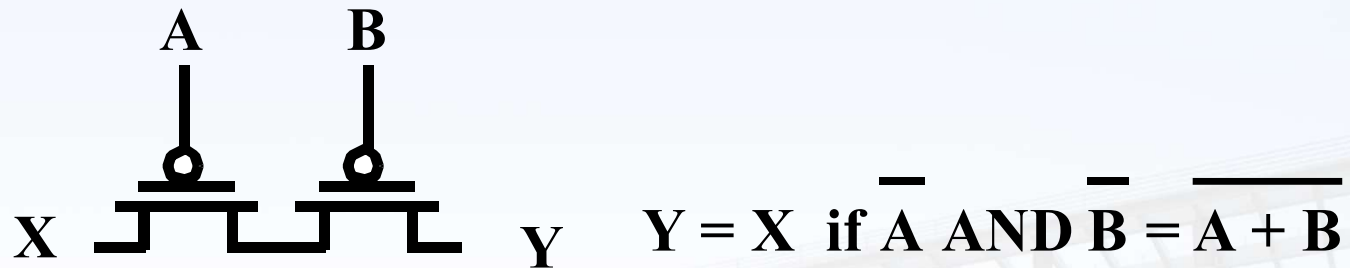
NMOS电路的串并联

NMOS器件的串联实现AND功能，并联实现OR功能



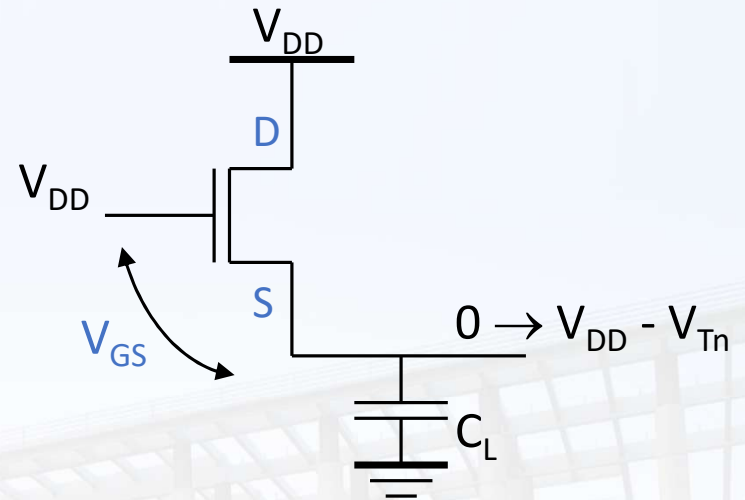
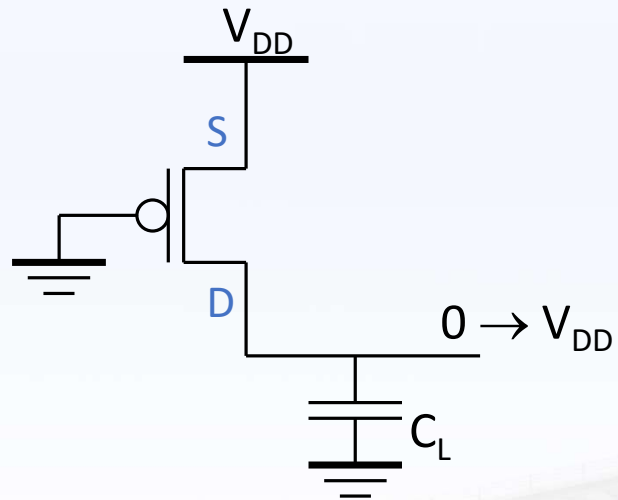
PMOS串并联电路

PMOS器件的串联与并联实现与NMOS互补功能

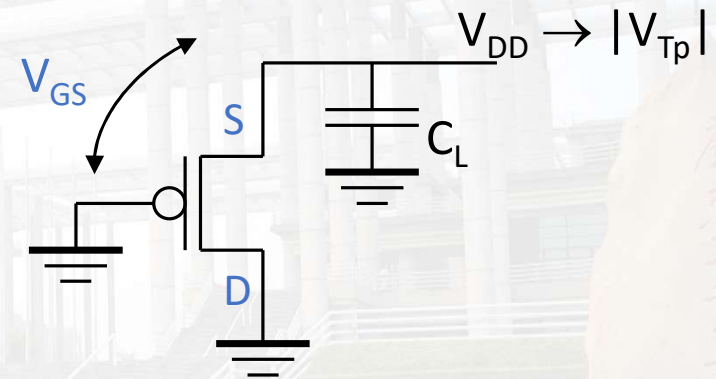
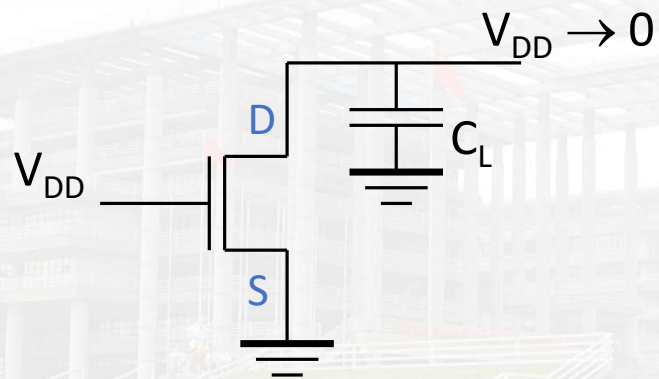


阈值问题

PUN



PDN



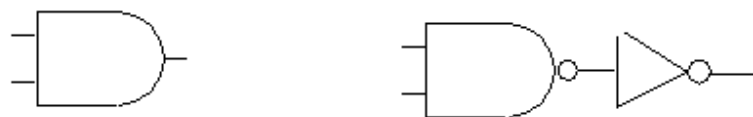
互补CMOS逻辑电路

PUN 是 **PDN**的补充:

$$\overline{A + B} = \bar{A}\bar{B}$$

$$\overline{\bar{A}\bar{B}} = \bar{A} + \bar{B}$$

互补逻辑:

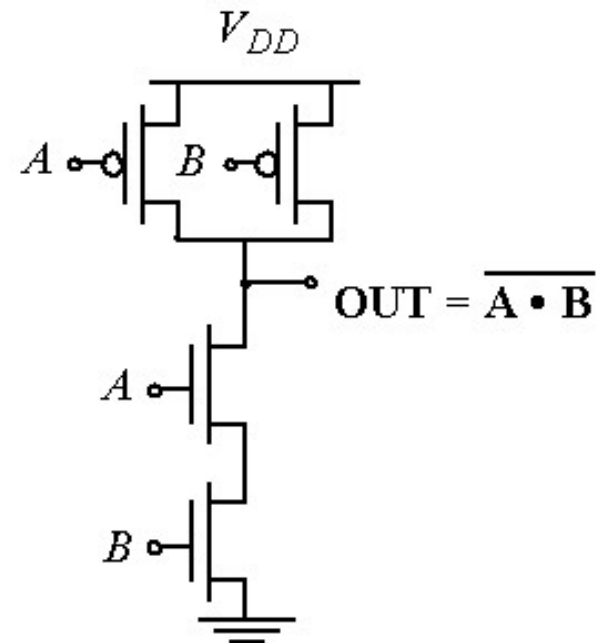


$$\text{AND} = \text{NAND} + \text{INV}$$

NAND

A	B	Out
0	0	1
0	1	1
1	0	1
1	1	0

Truth Table of a 2 input NAND gate



PDN: $G = A B \Rightarrow$ Conduction to GND

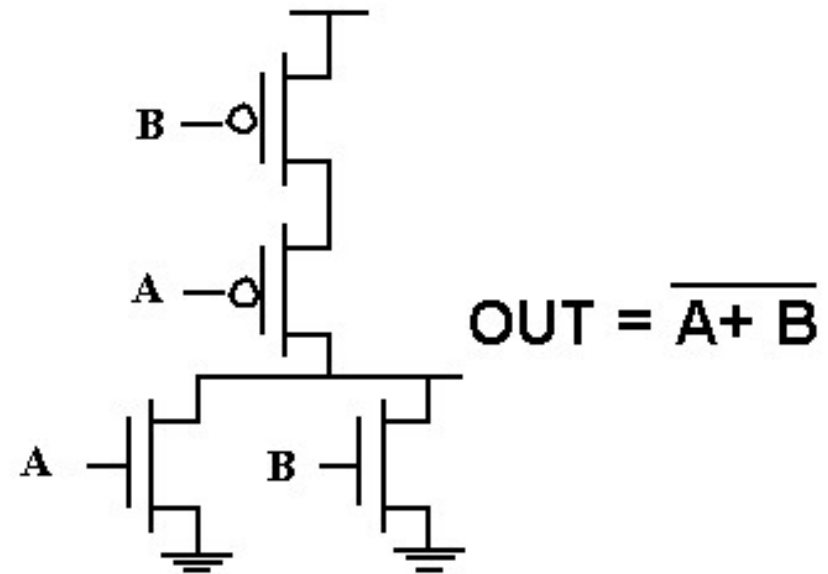
PUN: $F = \overline{A} + \overline{B} = \overline{AB} \Rightarrow$ Conduction to V_{DD}

$$G(In_1, In_2, In_3, \dots) \equiv F(\overline{In_1}, \overline{In_2}, \overline{In_3}, \dots)$$

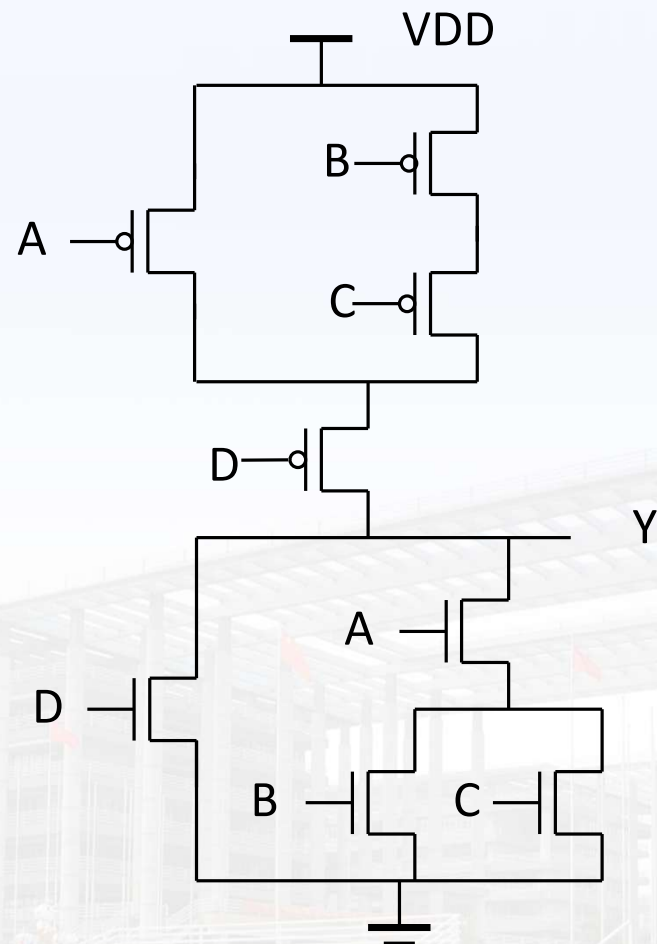
NOR

A	B	Out
0	0	1
0	1	0
1	0	0
1	1	0

Truth Table of a 2 input NOR gate

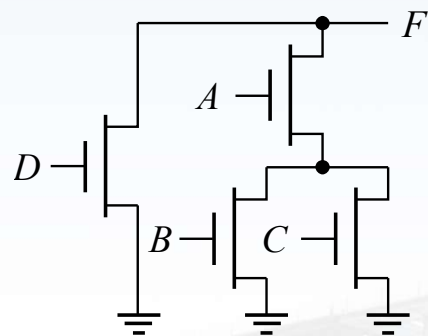


复杂CMOS门电路

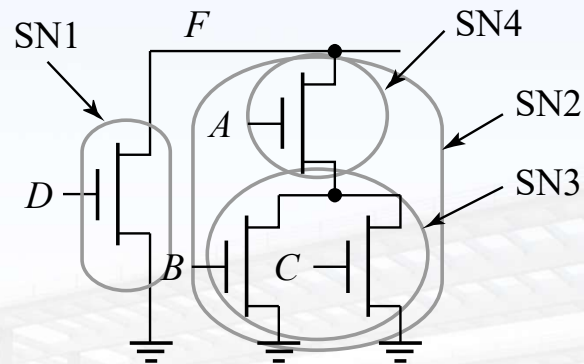


$$\text{OUT} = D + A \cdot (B + C)$$

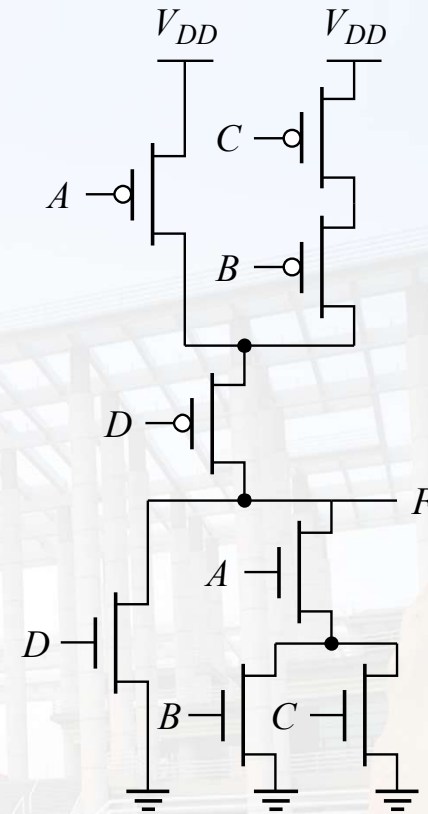
构建复杂逻辑单元过程



(a) pull-down network



(b) Deriving the pull-up network hierarchically by identifying sub-nets



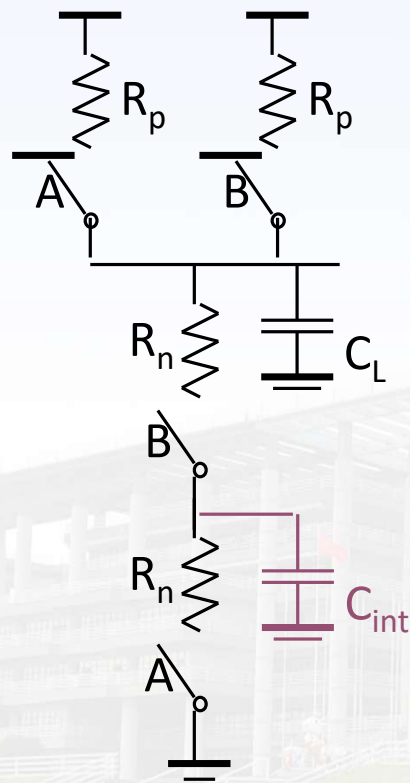
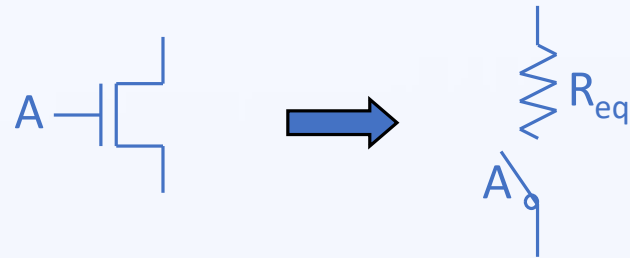
(c) complete gate



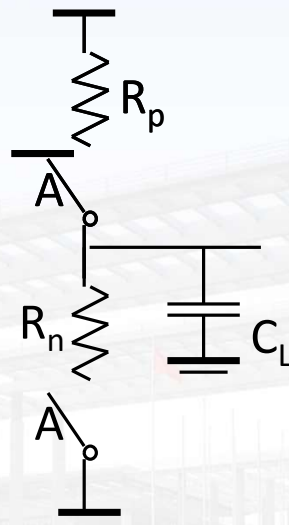
CMOS特点

- 无比电路, 具有最大的逻辑摆幅
- 在低电平状态不存在直流导通电流
- 静态功耗低
- 直流噪声容限大
- 采用对称设计获得最佳性能

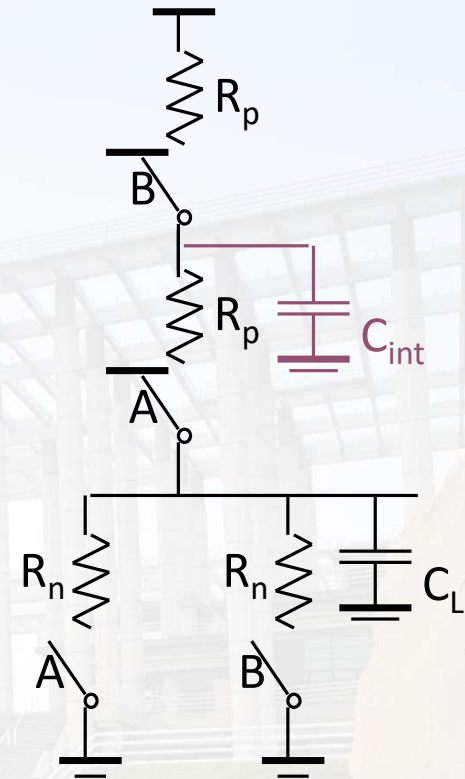
开关延迟模型



NAND2

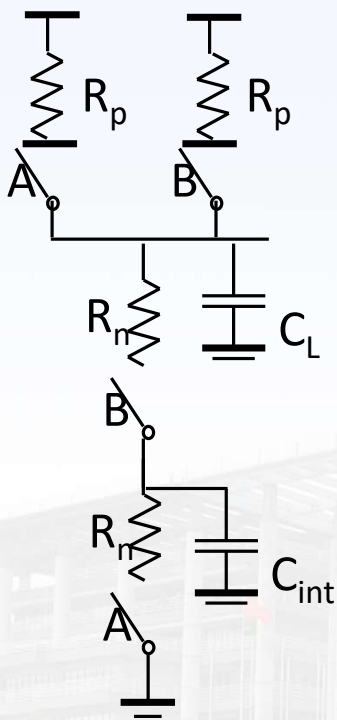


INV



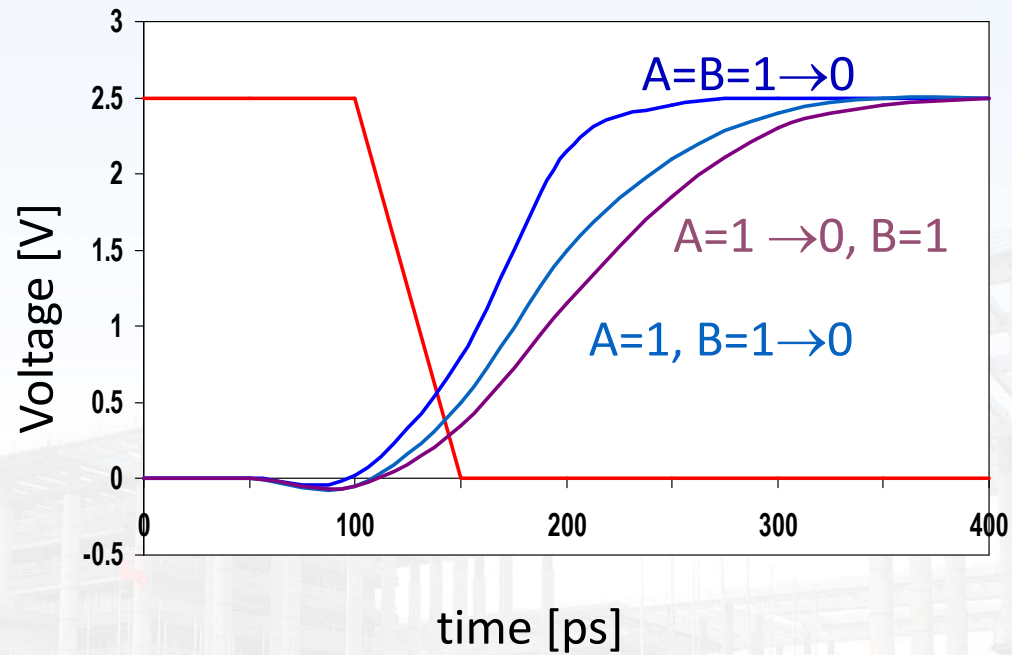
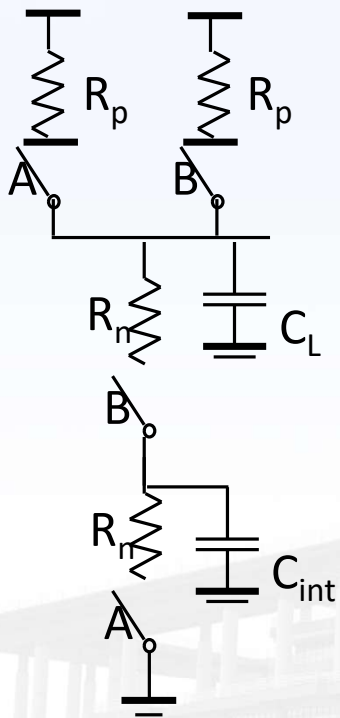
NOR2

延时特性



- 延时与输入信号相关
- L→H转换
 - 双输入均降低
 - 延时为 $0.69 R_p / 2 C_L$
 - 单输入降低
 - 延时为 $0.69 R_p C_L$
- H→L转换
 - 双输入均升高
 - 延时为 $0.69 2R_n C_L$

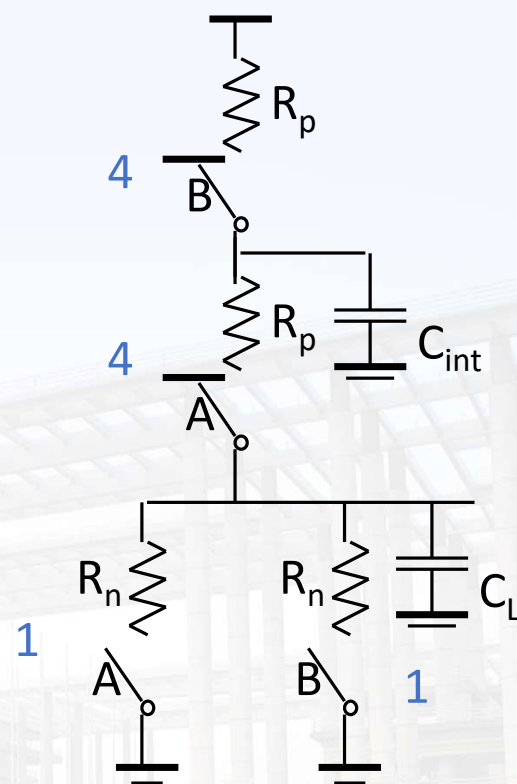
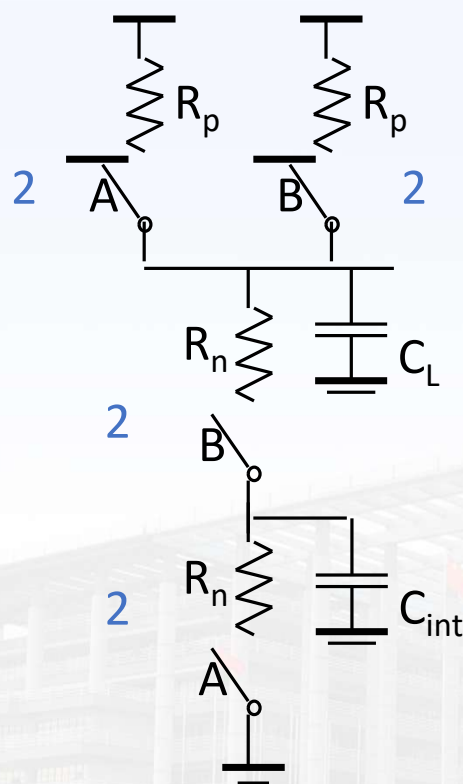
延时特性



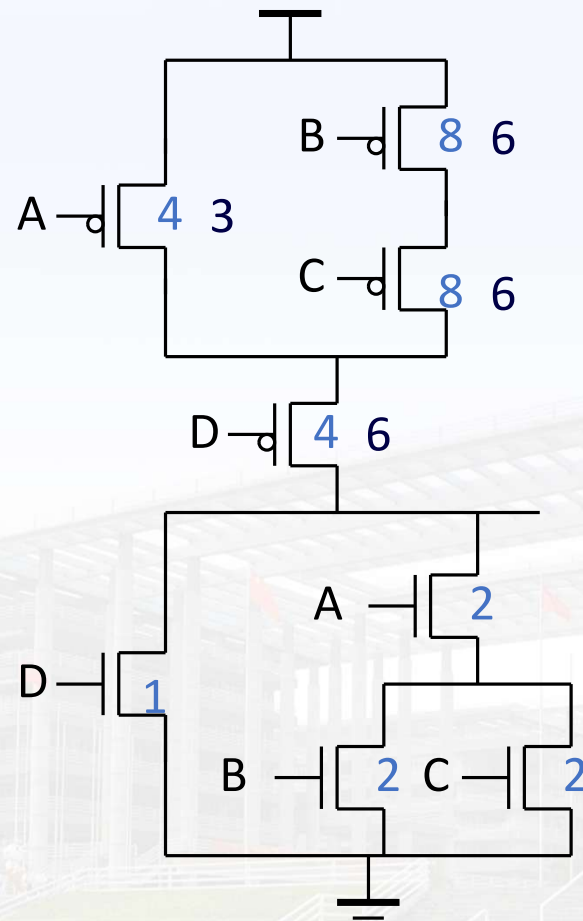
Input Data Pattern	Delay (psec)
A=B=0→1	67
A=1, B=0→1	64
A= 0→1, B=1	61
A=B=1→0	45
A=1, B=1→0	80
A= 1→0, B=1	81

NMOS = 0.5 μ m/0.25 μ m
 PMOS = 0.75 μ m/0.25 μ m
 C_L = 100 fF

晶体管尺寸

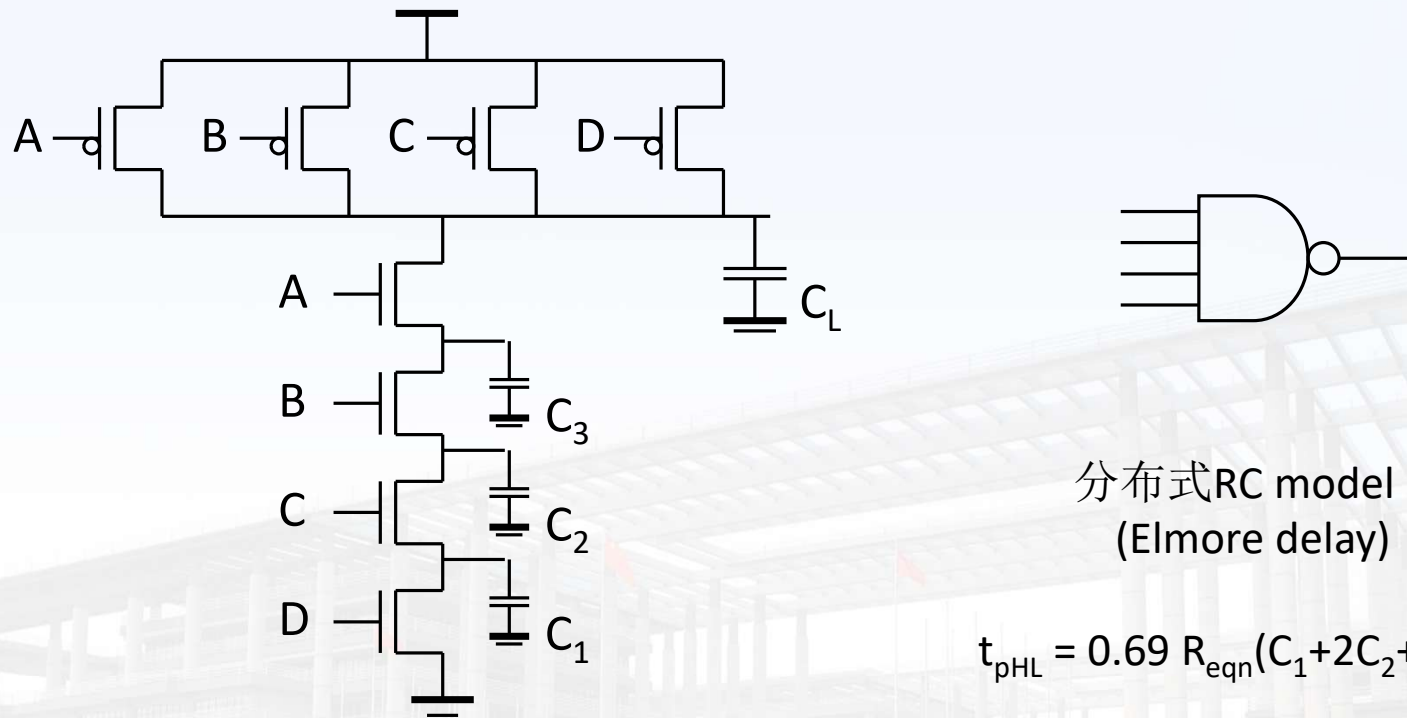


寄存器尺寸分析



$$OUT = D + A \cdot (B + C)$$

Fan-In分析

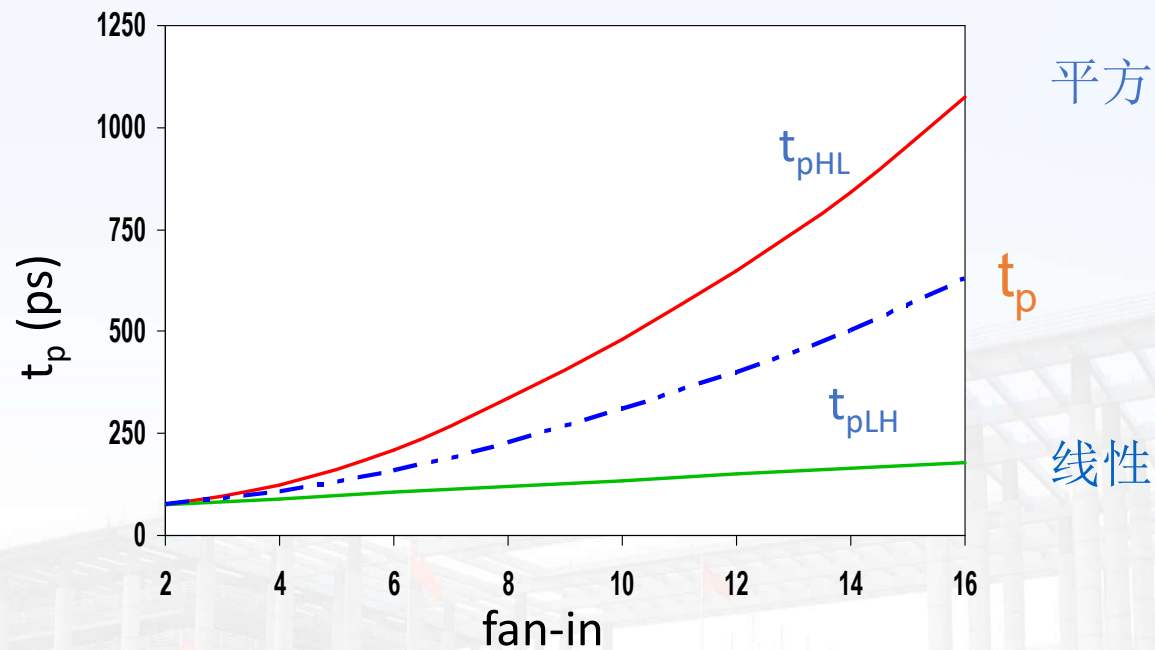


分布式RC model
(Elmore delay)

$$t_{pHL} = 0.69 R_{eqn}(C_1 + 2C_2 + 3C_3 + 4C_L)$$

传播延迟随着扇入量的增加而迅速恶化--在最坏的情况下呈四次方。

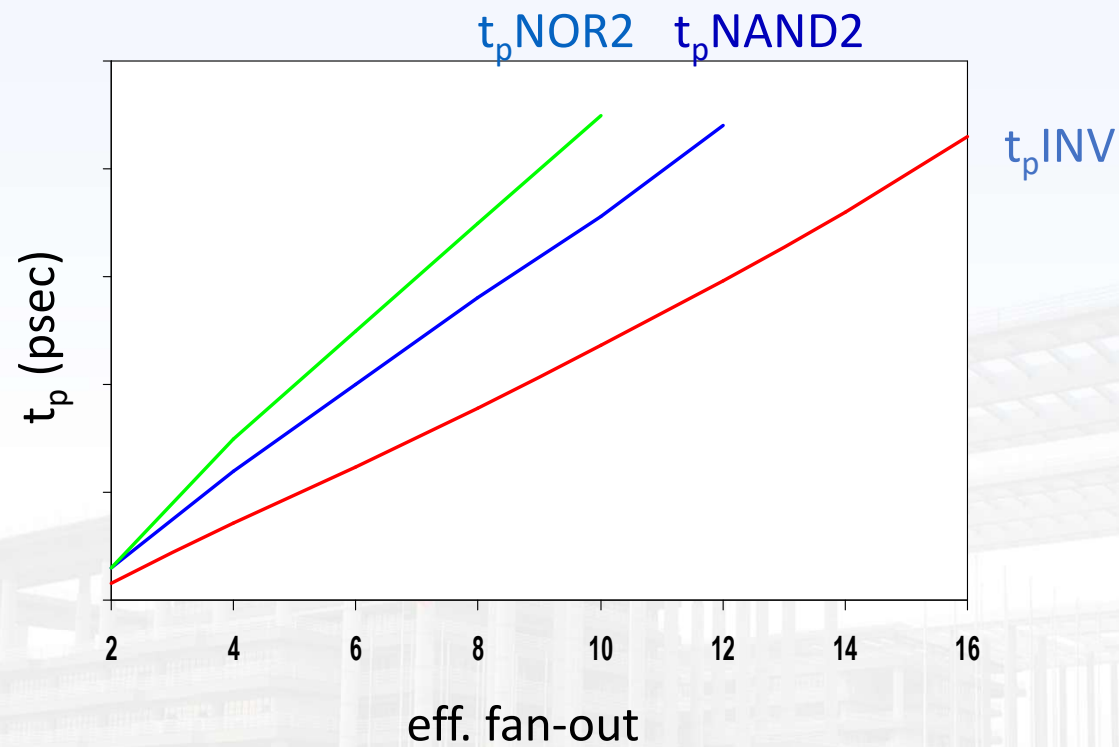
t_p 与 Fan-In 关系



在设计中应该尽可能避免4输入以上逻辑门



t_p 与 Fan-Out 关系



所有门有相同的驱动能力.

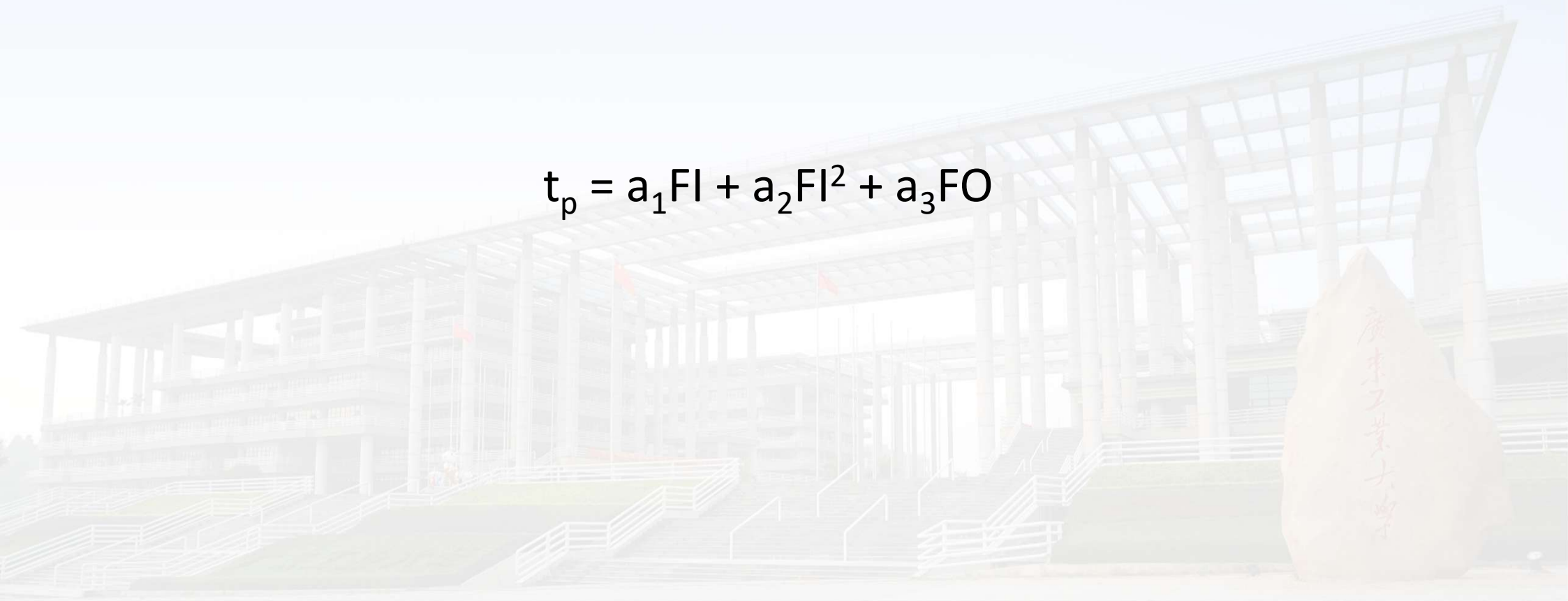
斜率是驱动能力的函数



Fan-In和Fan-Out

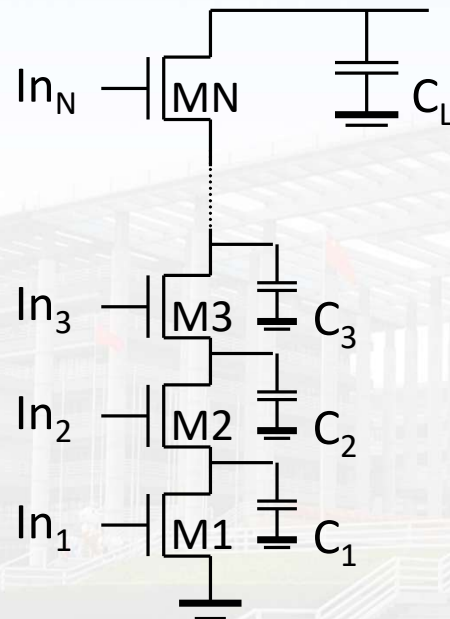
- Fan-in: 随着电阻和电容呈平方关系增长
- Fan-out: 每增加一个扇出的门增加两倍的删电容 C_L

$$t_p = a_1 FI + a_2 FI^2 + a_3 FO$$



优化设计技术

- 调整晶体管尺寸
 - 但同时会增加扇出的电容
- 逐级调整晶体管尺寸



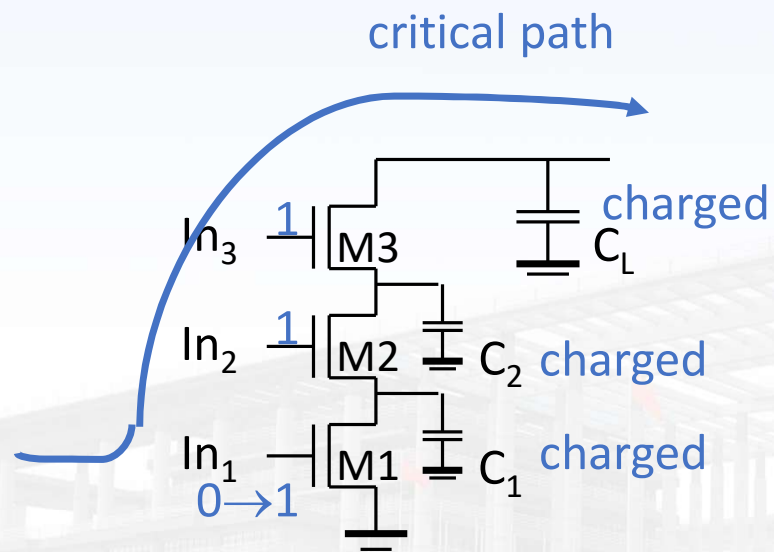
分布式RC传输线

$$M1 > M2 > M3 > \dots > MN$$

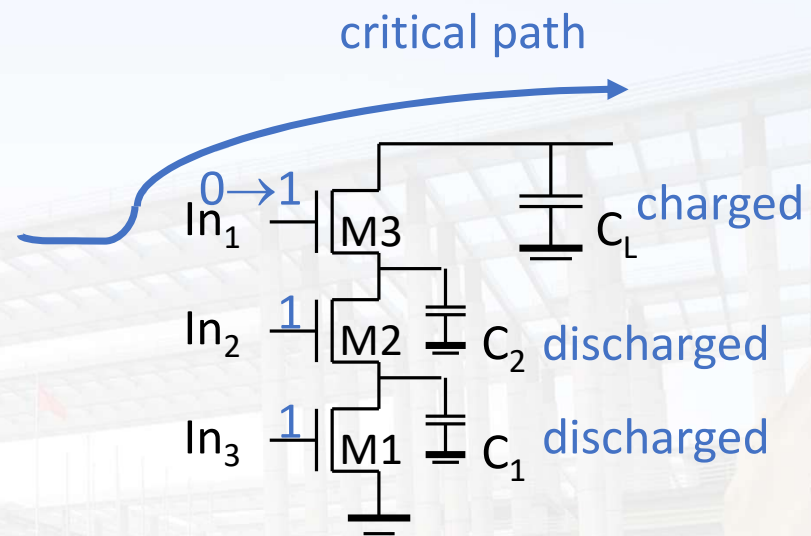
降低了器件的等效电阻

优化设计技术

• 门重排



延时由电容 C_L, C_1, C_2 决定

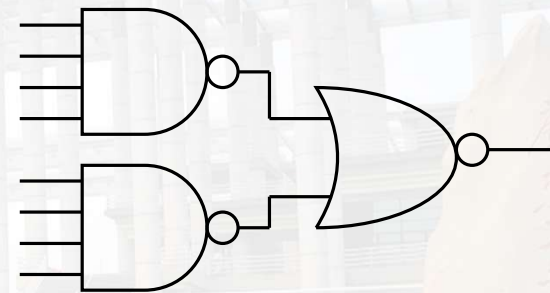
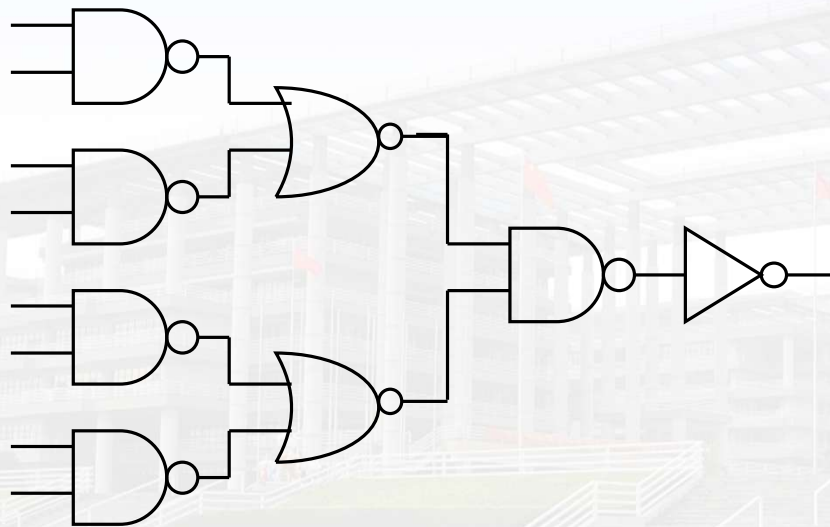
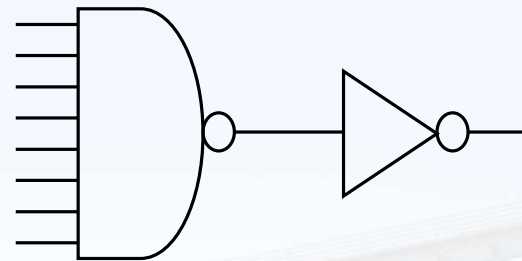


延时由电容 C_L 决定

优化设计技术

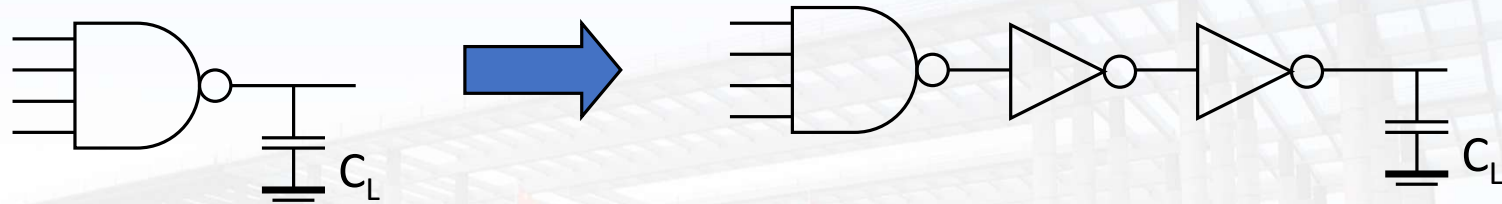
- 优化逻辑结构

$$F = ABCDEFGH$$



优化设计技术

- 利用Buffer单元隔离扇入和扇出





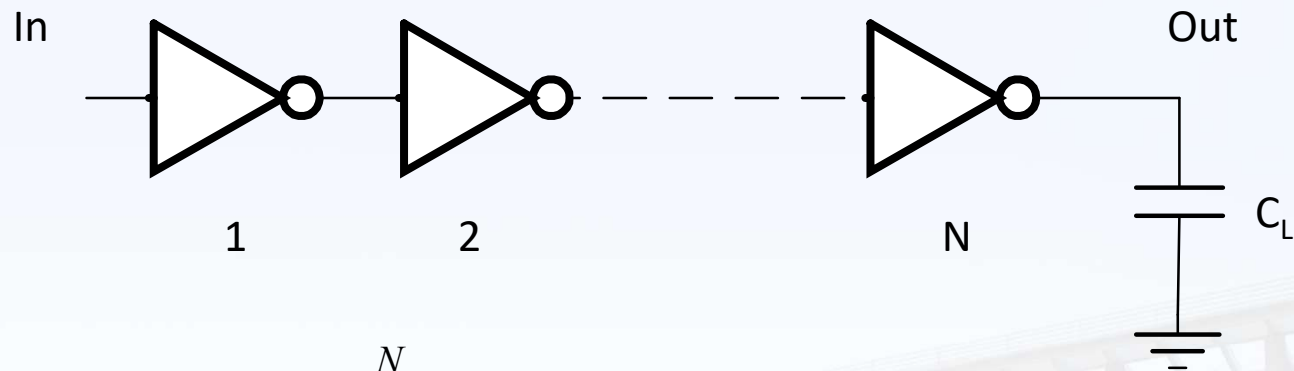
优化设计技术

- 减小电压摆幅

$$\begin{aligned} t_{pHL} &= 0.69 (3/4 (C_L V_{DD}) / I_{DSATn}) \\ &= 0.69 (3/4 (C_L V_{swing}) / I_{DSATn}) \end{aligned}$$

- 线性减少延时
- 降低系统功耗
- 但是后续单元的延时会增加!
- 需要在后续单元中增加灵敏放大器来恢复小信号（存储器单元）

缓存单元



$$Delay = \sum_{i=1}^N (p_i + g_i \cdot f_i) \quad (\text{in units of } \tau_{inv})$$

给定 N : $C_{i+1}/C_i = C_i/C_{i-1}$
查找 N : $C_{i+1}/C_i \sim 4$



逻辑努力

$$\begin{aligned} Delay &= k \cdot R_{unit} C_{unit} \left(1 + \frac{C_L}{\gamma C_{in}} \right) \\ &= \tau(p + g \cdot f) \end{aligned}$$

p – 本征延时($3kR_{unit}C_{unit}\gamma$) 的比值
g – 逻辑努力 ($kR_{unit}C_{unit}$)
f – 等效扇出

以单元反相器作为基准: $g_{inv} = 1, p_{inv} = 1$



逻辑努力

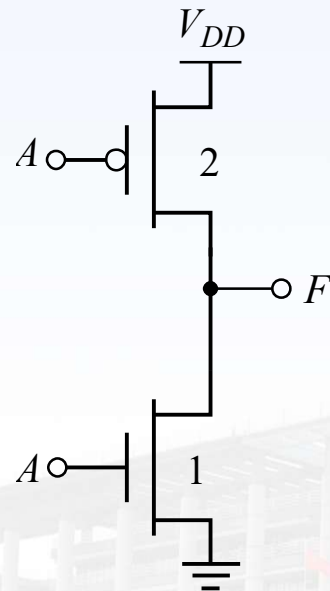
- 基准的确立：反相器是静态CMOS电路中最小的逻辑努力和本证延时单元
- 定义：一个门的逻辑努力代表在相同的电流下输入电容与反相器的比值。
- 随着逻辑门结构复杂化，逻辑努力也会随之增加。

逻辑努力

逻辑努力:

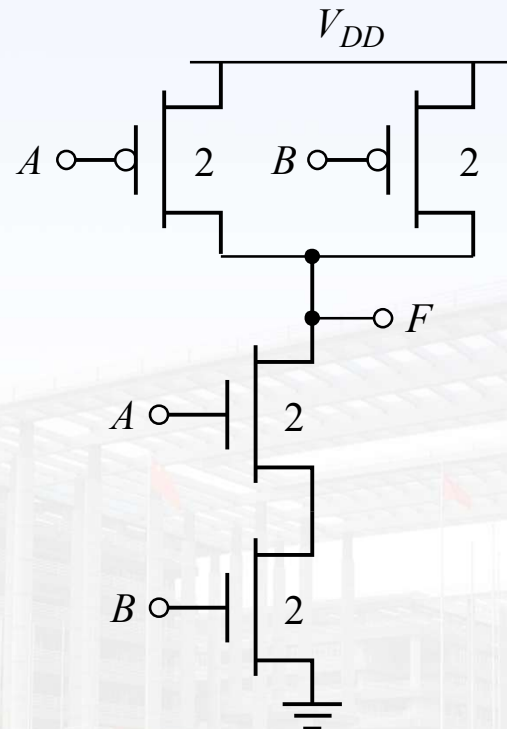
前提: 一个门与一个反向器提供相同的输出电流

内涵: 它所表现出的输入电容比反相器大多少



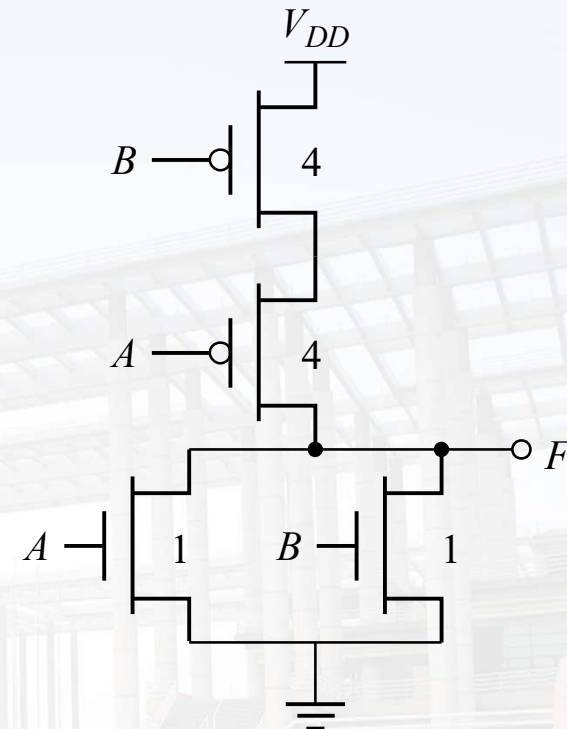
Inverter

$$g = 1$$



2-input NAND

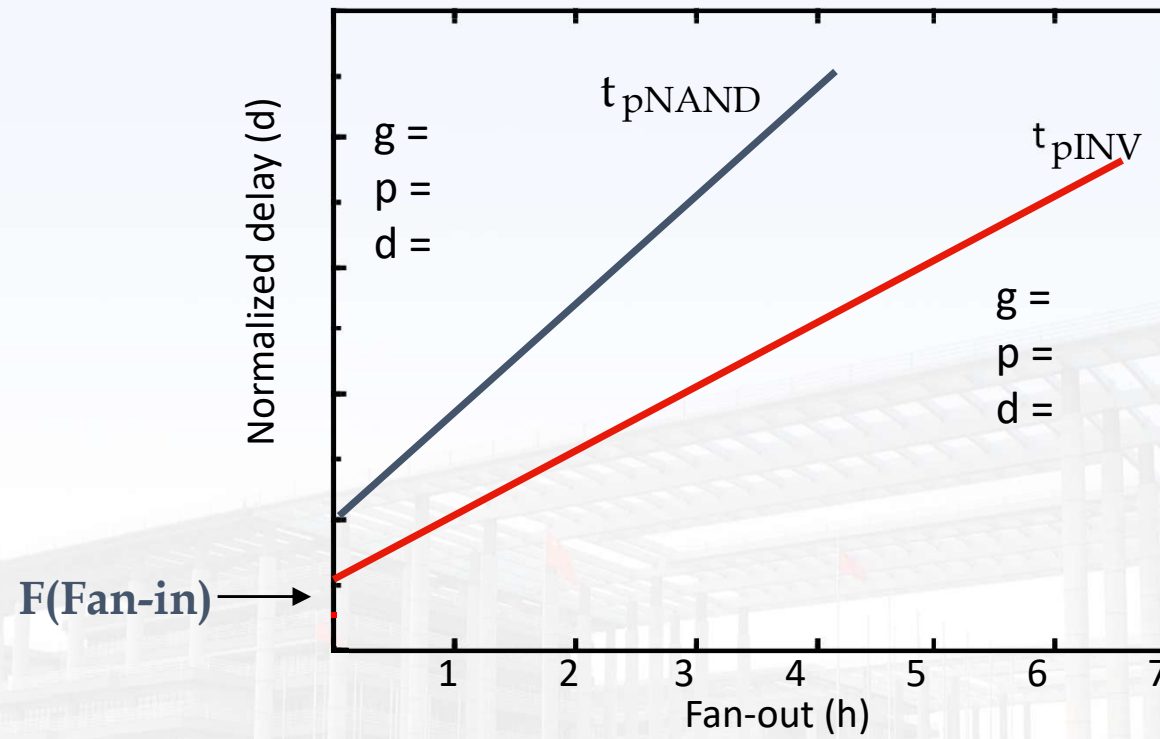
$$g = 4/3$$



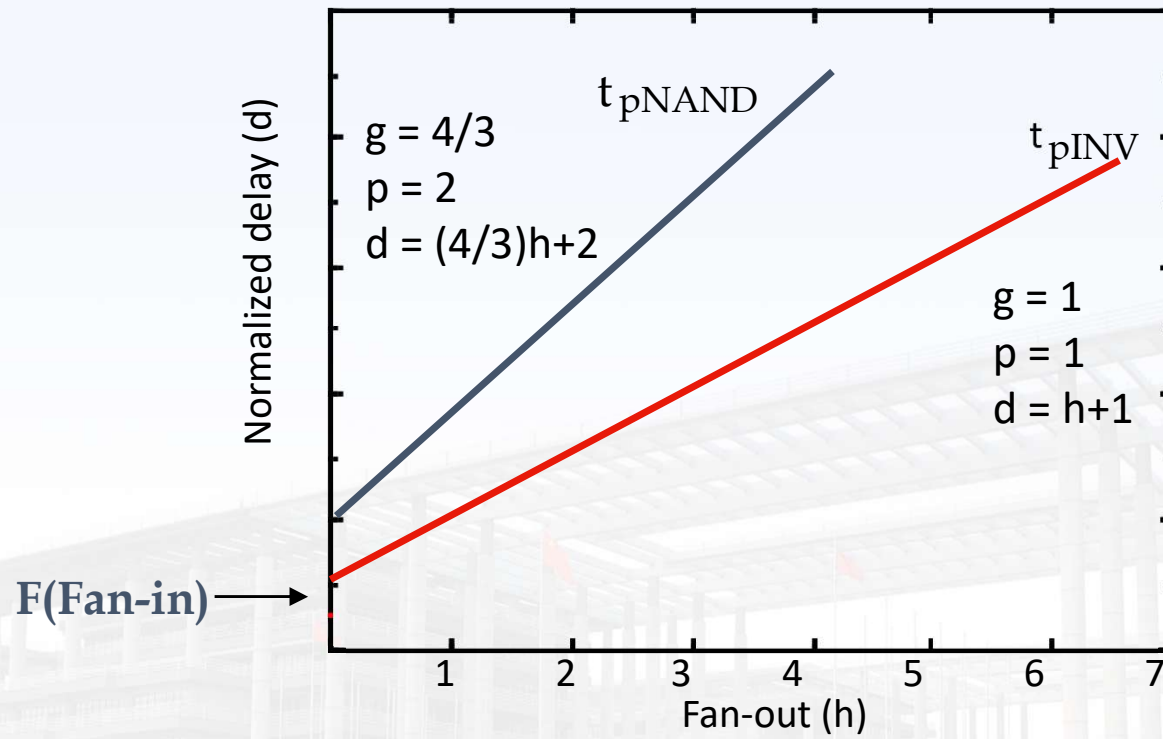
2-input NOR

$$g = 5/3$$

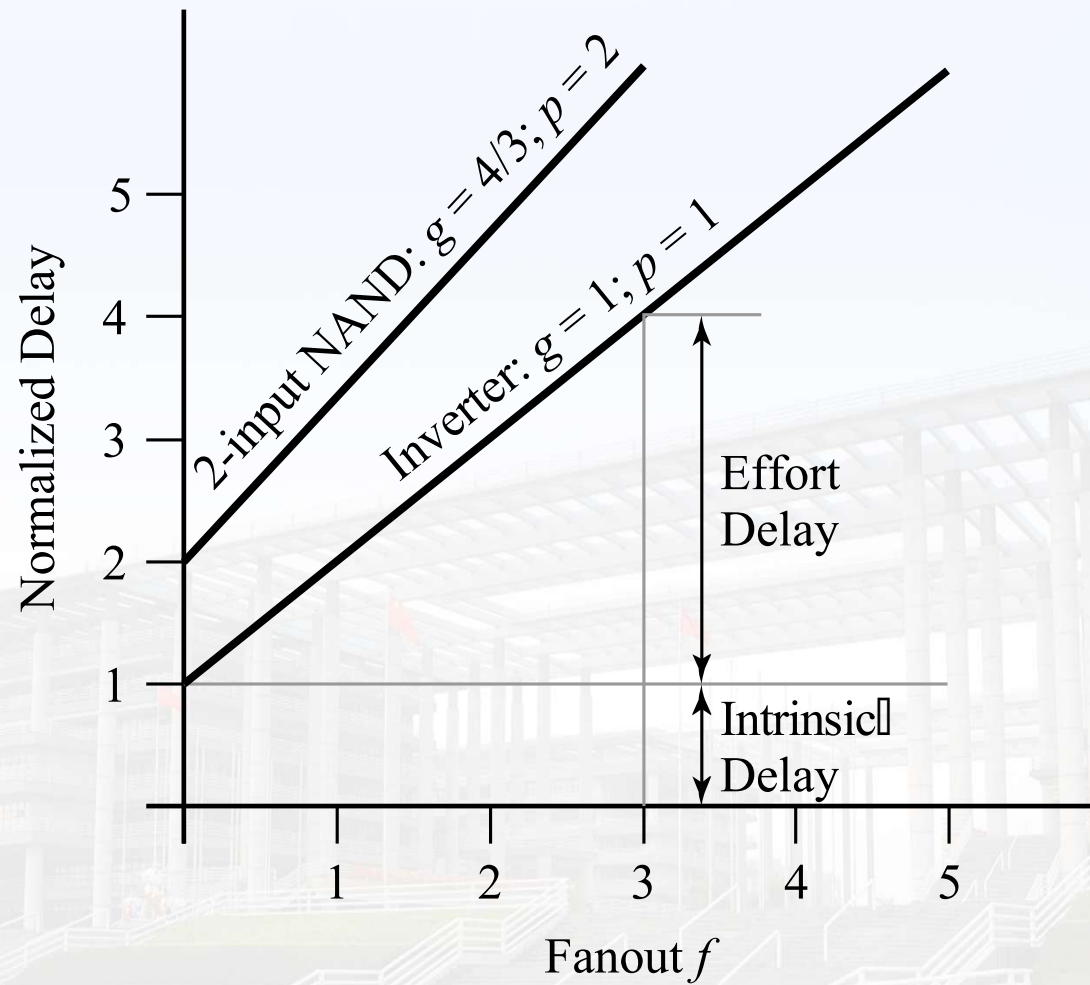
逻辑努力



逻辑努力



逻辑努力





逻辑努力

Gate Type	Number of Inputs			
	1	2	3	n
Inverter	1			
NAND		$4/3$	$5/3$	$(n + 2)/3$
NOR		$5/3$	$7/3$	$(2n + 1)/3$
Multiplexer		2	2	2
XOR		4	12	

功耗分析

- 静态CMOS电路中动态功耗主要取决于逻辑翻转概率，仅0-1翻转产生。

$$\alpha_{0 \rightarrow 1} = p_0 \cdot p_1 = p_0 \cdot (1 - p_0)$$

N输入逻辑门

$$\alpha_{0 \rightarrow 1} = \frac{N_0}{2^N} \cdot \frac{N_1}{2^N} = \frac{N_0 \cdot (2^N - N_0)}{2^{2N}}$$

NOR翻转功耗分析

- 逻辑表示:

$$Y = \overline{(A|B)}$$

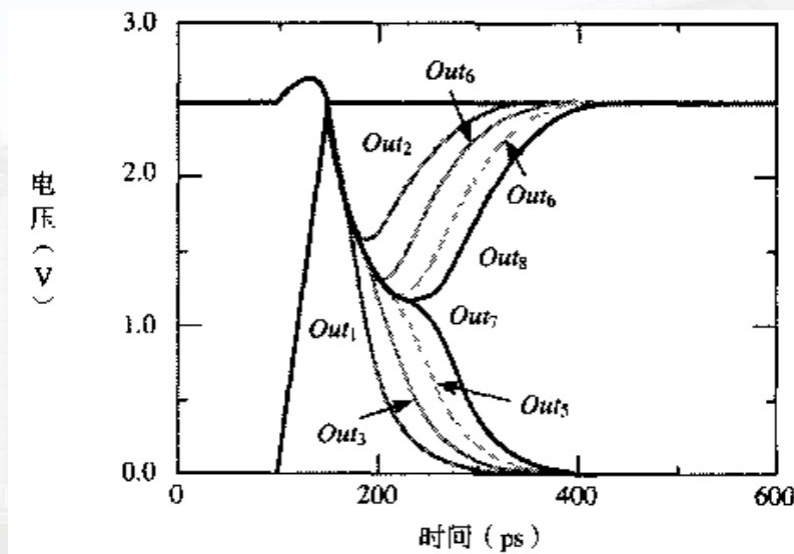
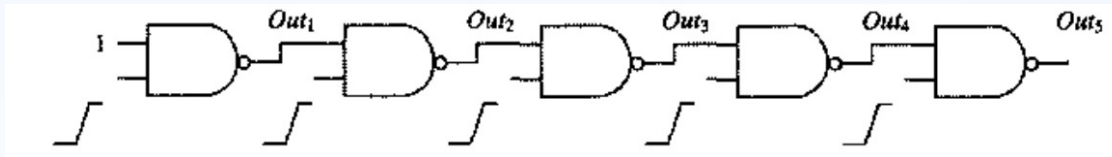
真值表:

A	B	Y
0	0	1
0	1	0
1	0	0
1	1	0

$$\alpha_{0 \rightarrow 1} = \frac{N_0 \cdot (2^N - N)}{2^{2N}} = \frac{3 \cdot (2^2 - 3)}{2^{2 \cdot 2}} = \frac{3}{16}$$

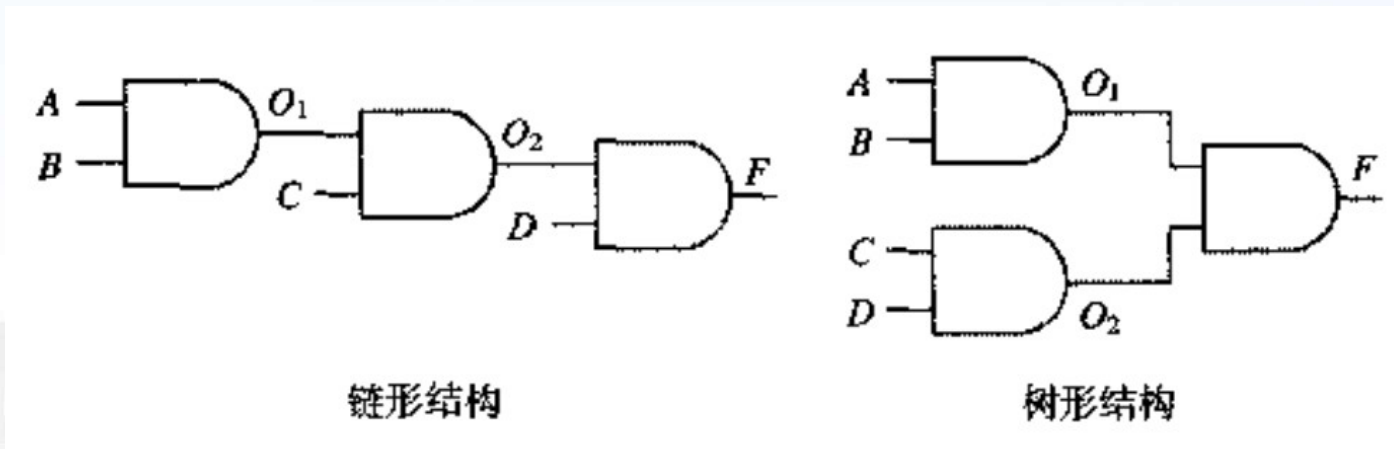
动态或虚假功耗

- 存在信号间延时时会造成多次翻转。
- 多逻辑单元级联时：



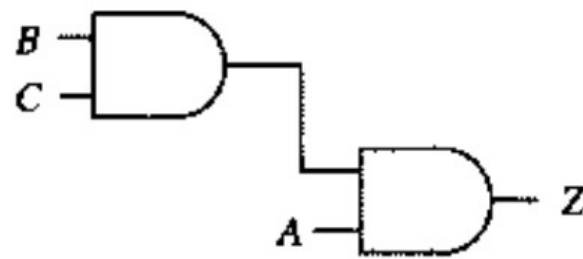
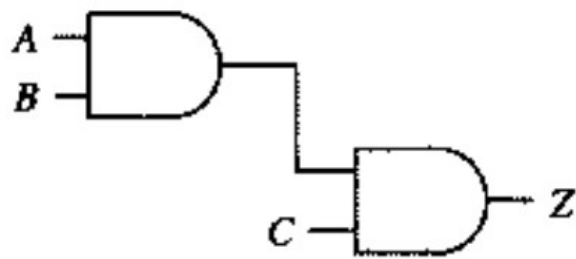
降低功耗方法

- 逻辑重组
 - 通过改变逻辑拓扑结构，降低门翻转概率



降低功耗方法

- 输入重排
 - 通过分析输入节点的0/1概率分布来优化电路设计



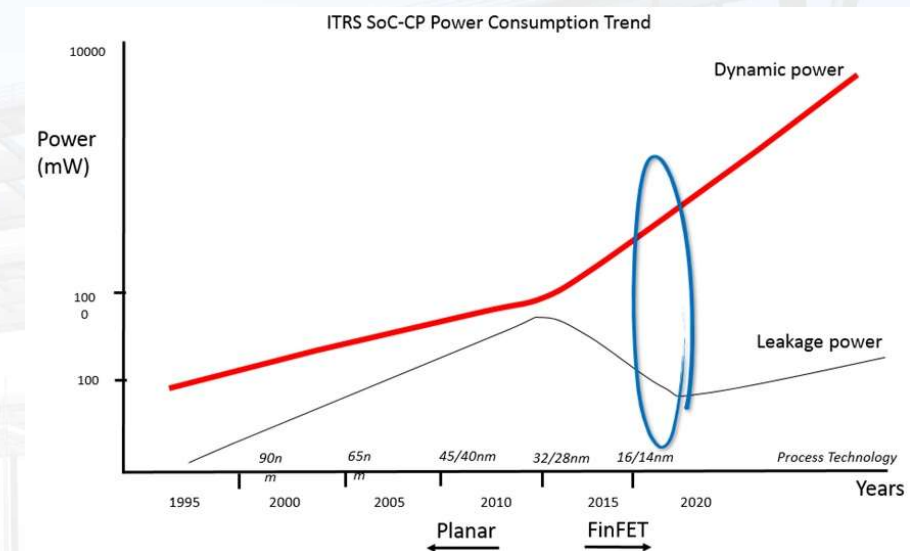
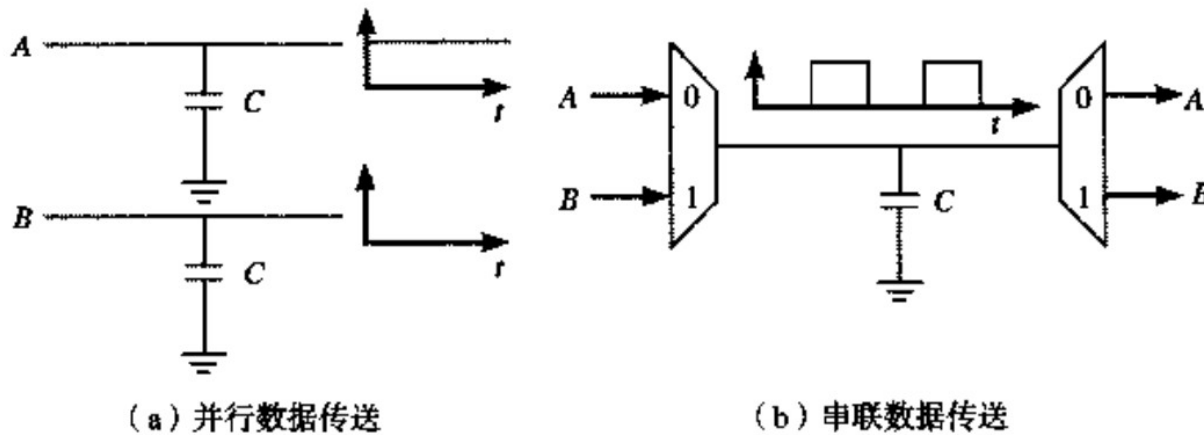
$$P(A=1) = 0.5$$

$$P(B=1) = 0.2$$

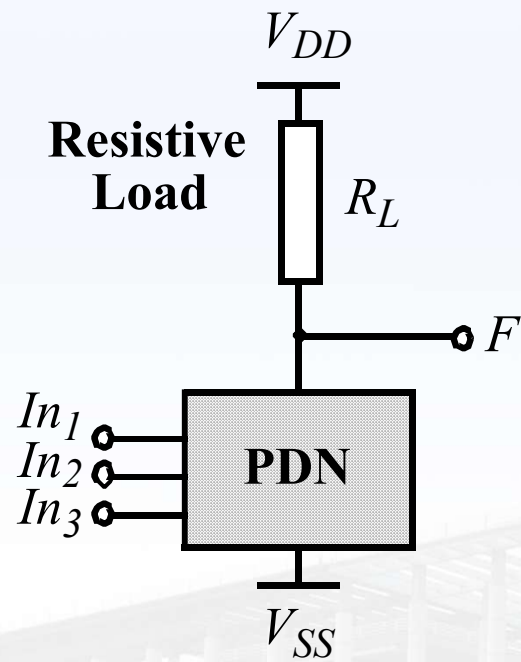
$$P(C=1) = 0.1$$

降低功耗方法

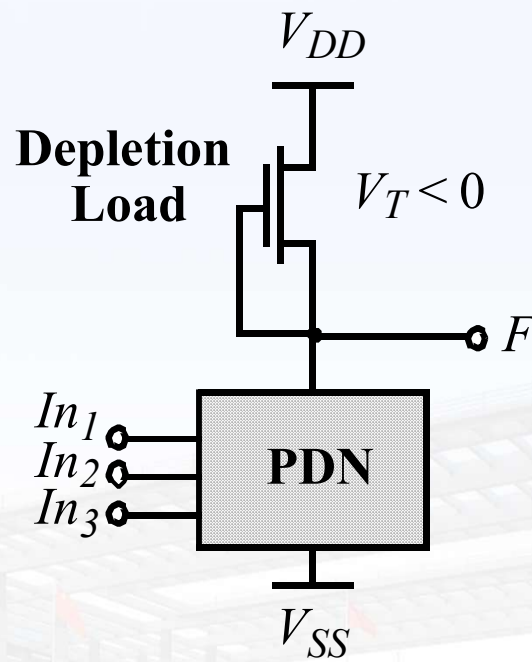
- 分时资源复用
 - 分时复用硬件资源，降低硬件电路面积



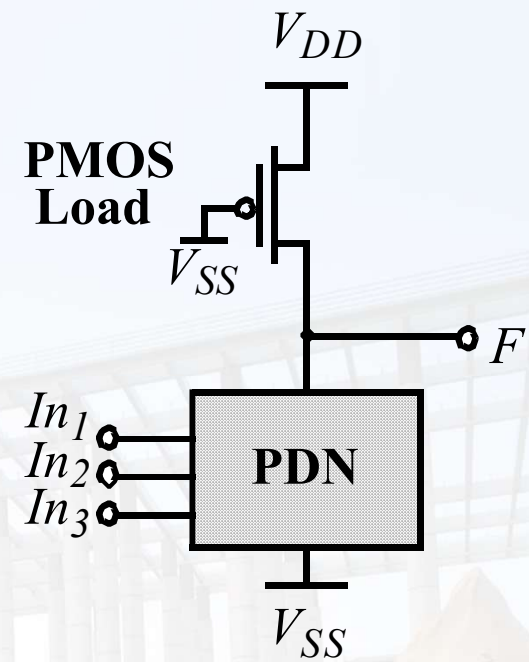
有比逻辑



(a) resistive load



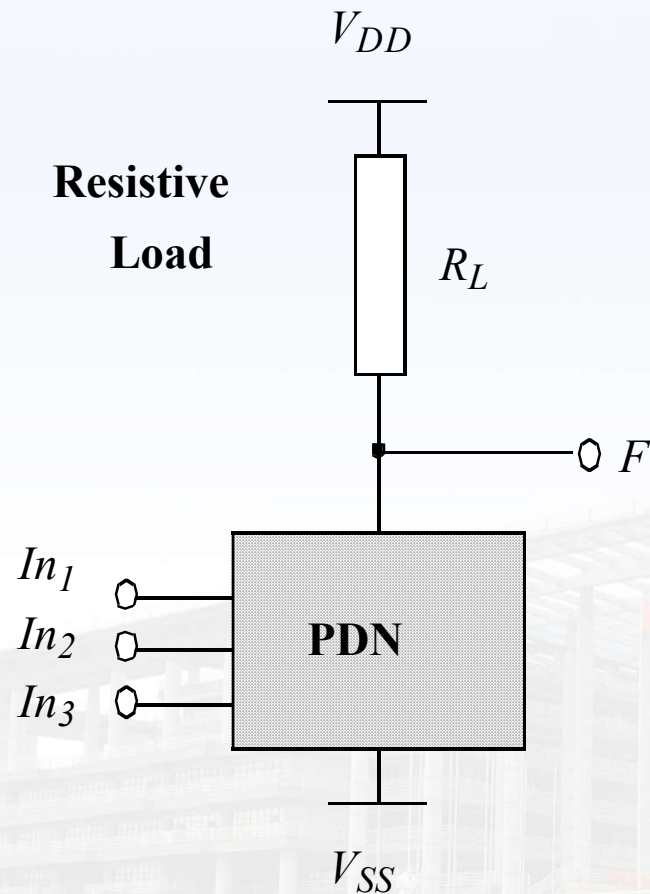
(b) depletion load NMOS



(c) pseudo-NMOS

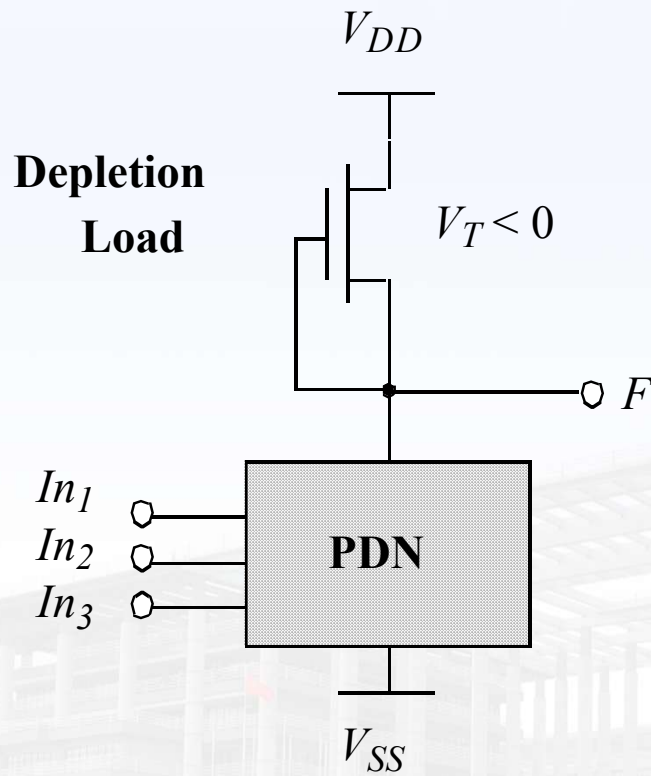
目标：减少互补CMOS电路结构中的MOS器件数目

有比逻辑

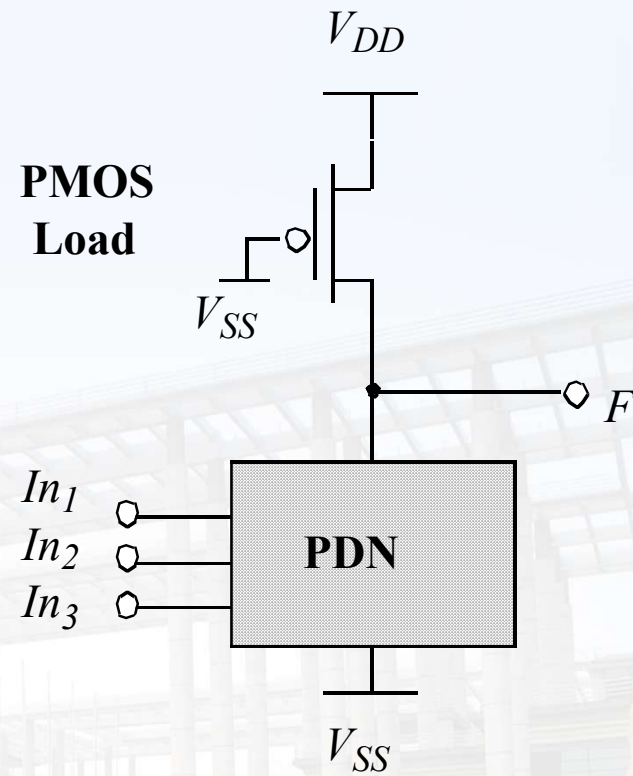


- N个晶体管+1个负载
- 高电平等于电源电压
- 低电平为等效电阻的比值
- 对称的响应
- 存在静态功耗
- $t_{pL} = 0.69R_L C_L$

有源负载

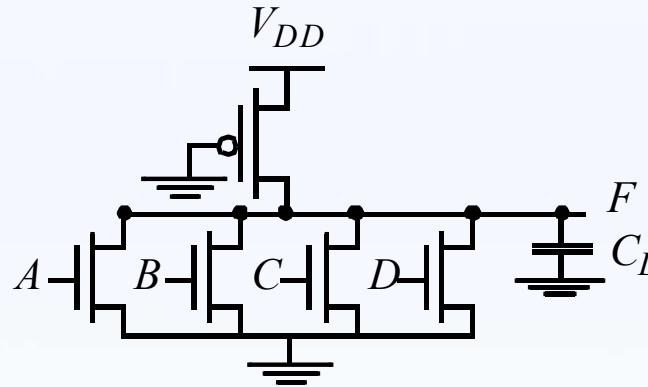


depletion load NMOS



pseudo-NMOS

伪NMOS



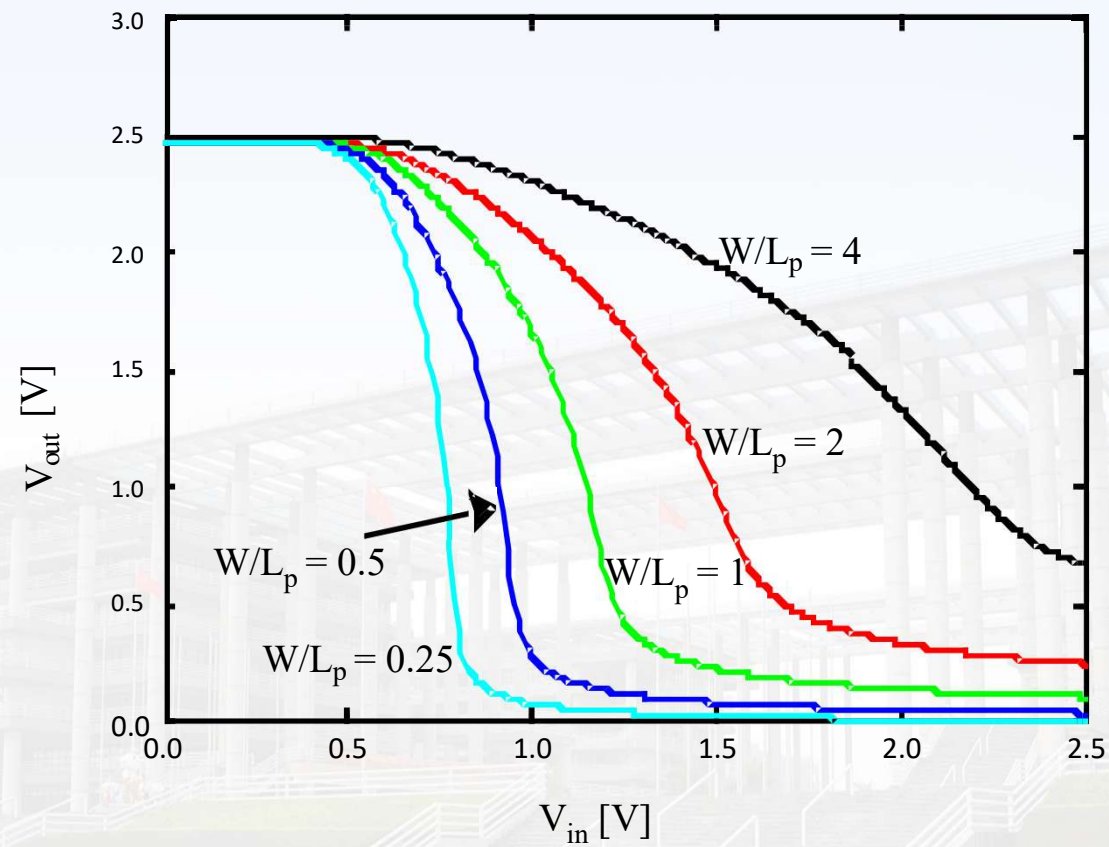
$V_{OH} = V_{DD}$ (similar to complementary CMOS)

$$k_n \left((V_{DD} - V_{Tn}) V_{OL} - \frac{V_{OL}^2}{2} \right) = \frac{k_p}{2} (V_{DD} - |V_{Tp}|)^2$$

$$V_{OL} = (V_{DD} - V_T) \left[1 - \sqrt{1 - \frac{k_p}{k_n}} \right] \text{ (assuming that } V_T = V_{Tn} = |V_{Tp}| \text{)}$$

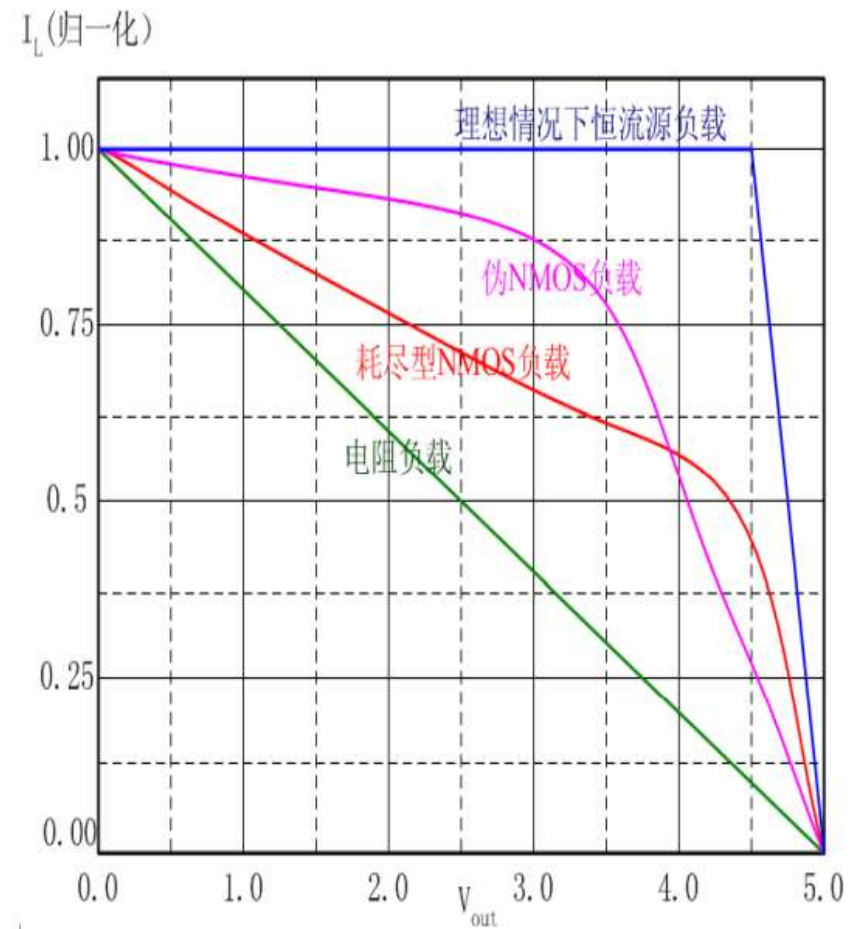
- 优势：更小的面积和负载。
- 不足：静态功耗

伪NMOS 电压传输曲线

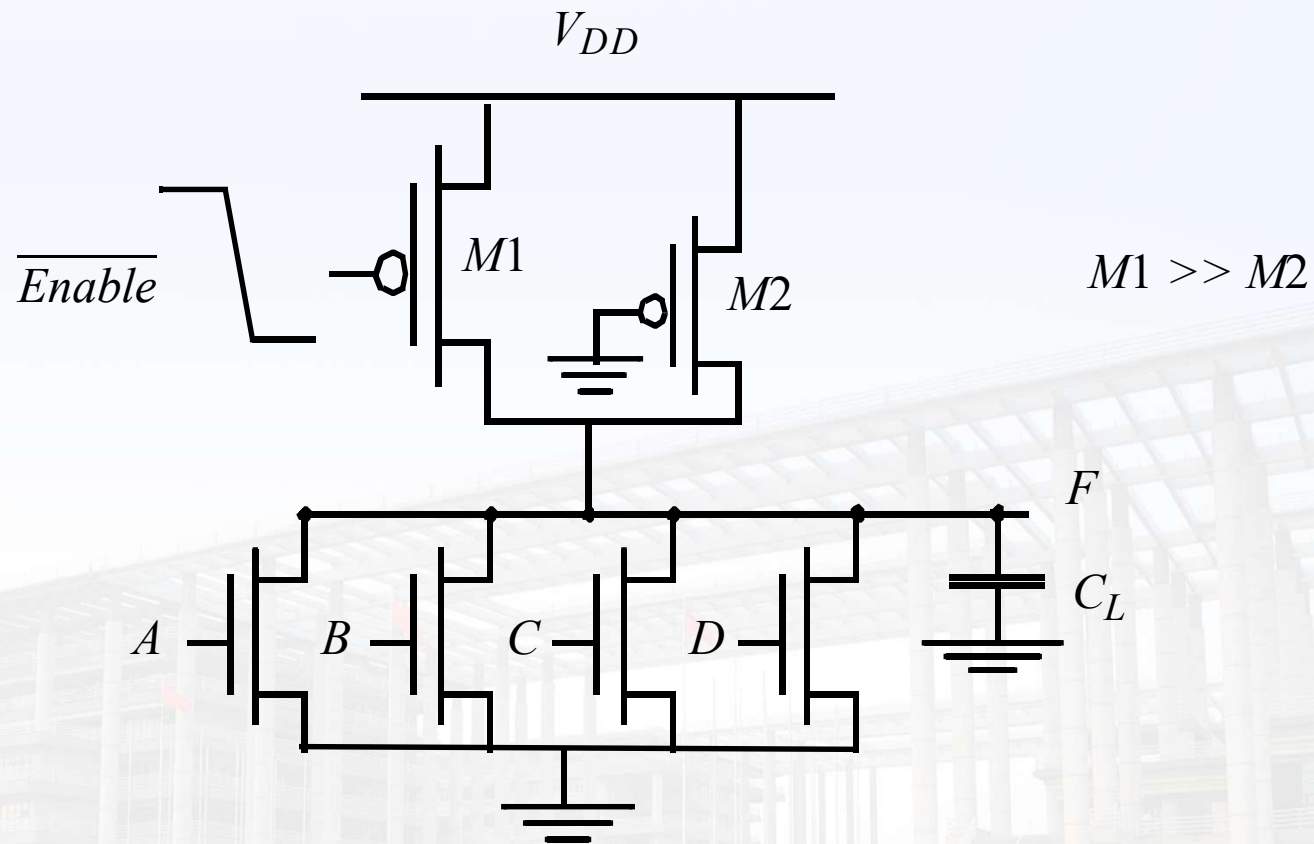


电压传输曲线

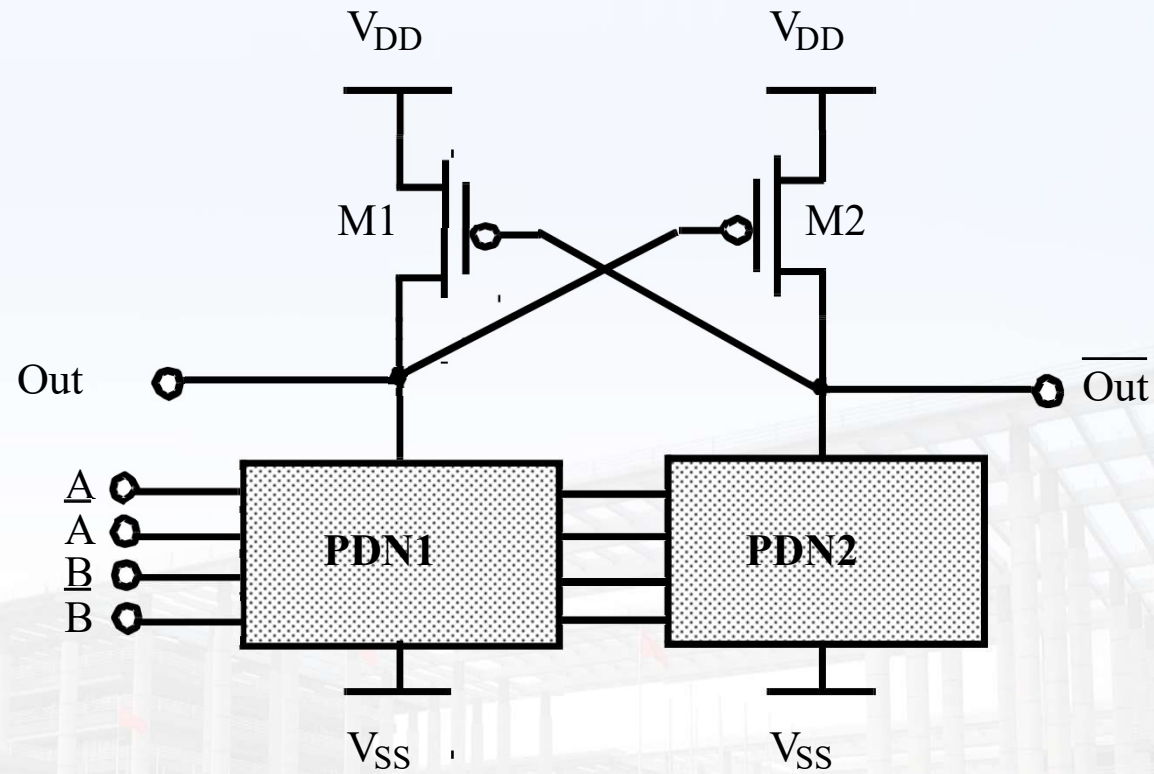
$$I_L = \frac{V_{DD} - V_{out}}{R_L}$$



更好的负载器件

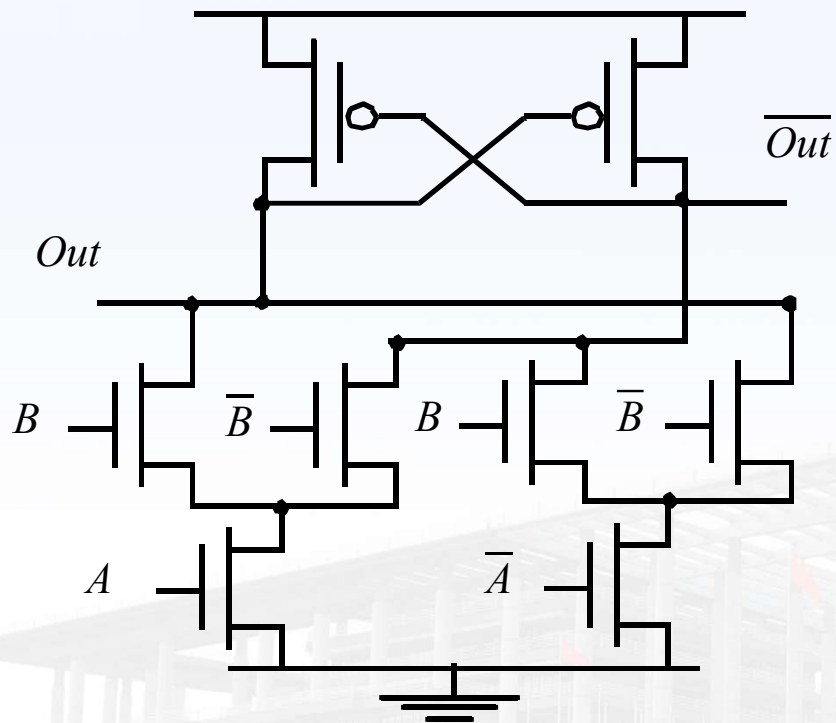


更好的负载器件

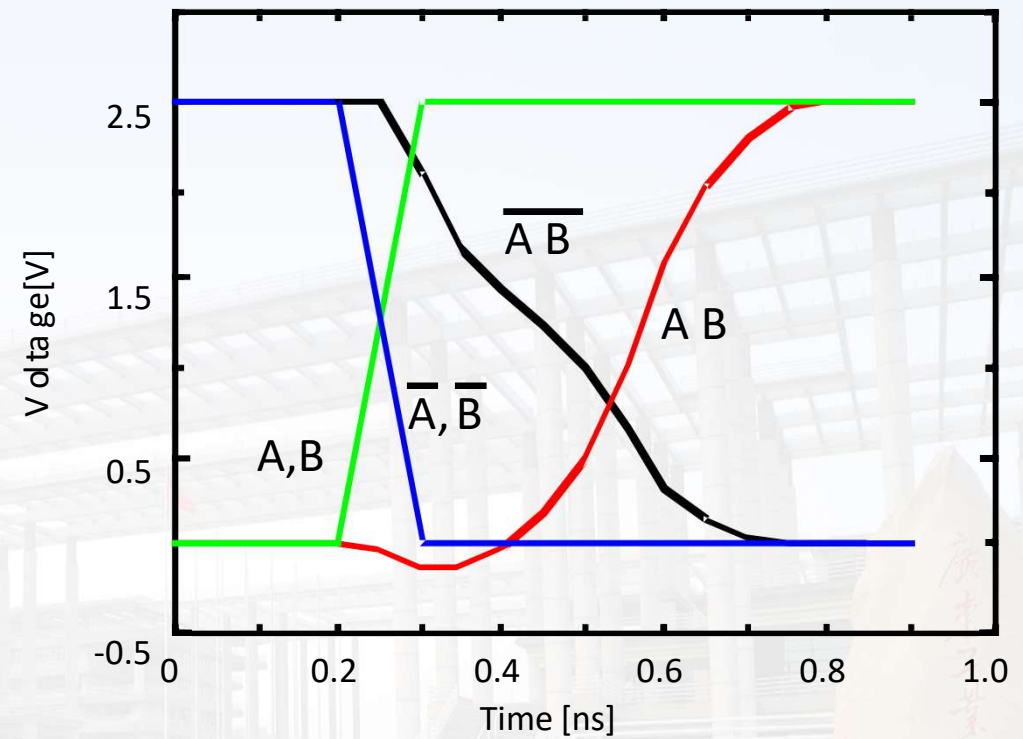


Differential Cascode Voltage Switch Logic (DCVSL)

DCVSL 案例

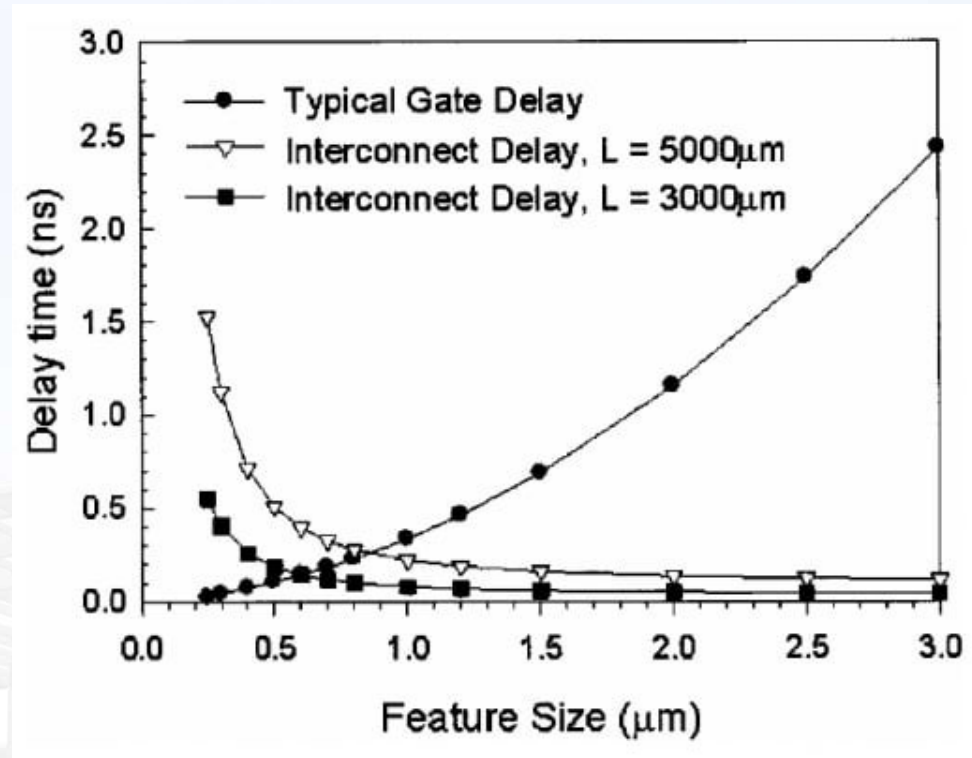


XOR-NXOR gate

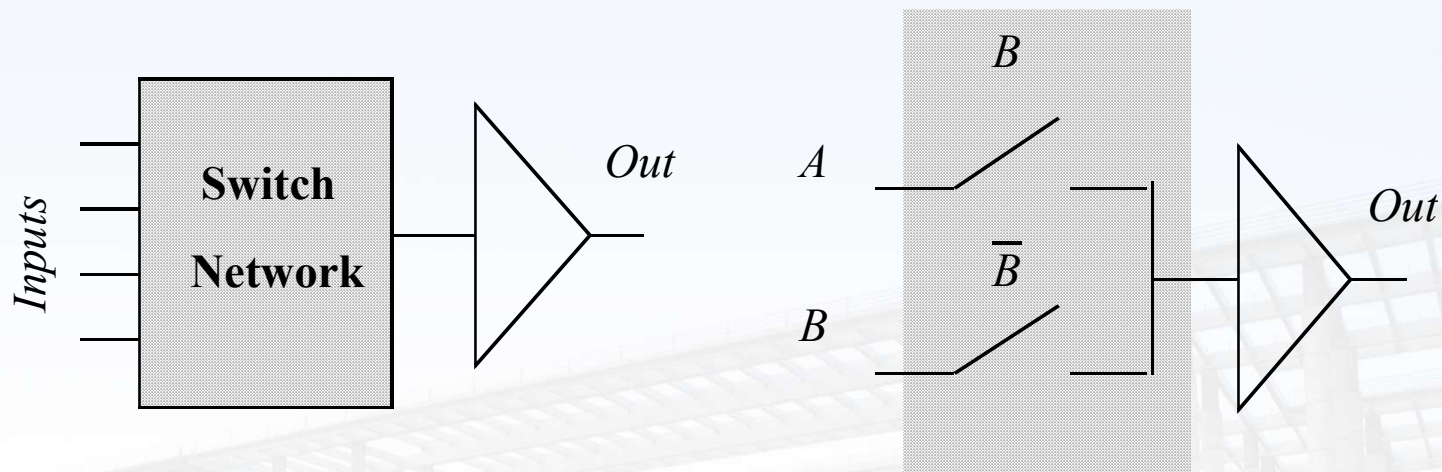


DCVSL 门特性

- 优势：
 - 提供差分输出
 - 硬件资源优势
 - 差分输出时差小
 - 轨对轨摆幅
- 缺点：
 - 布线资源增多
 - 动态功耗大

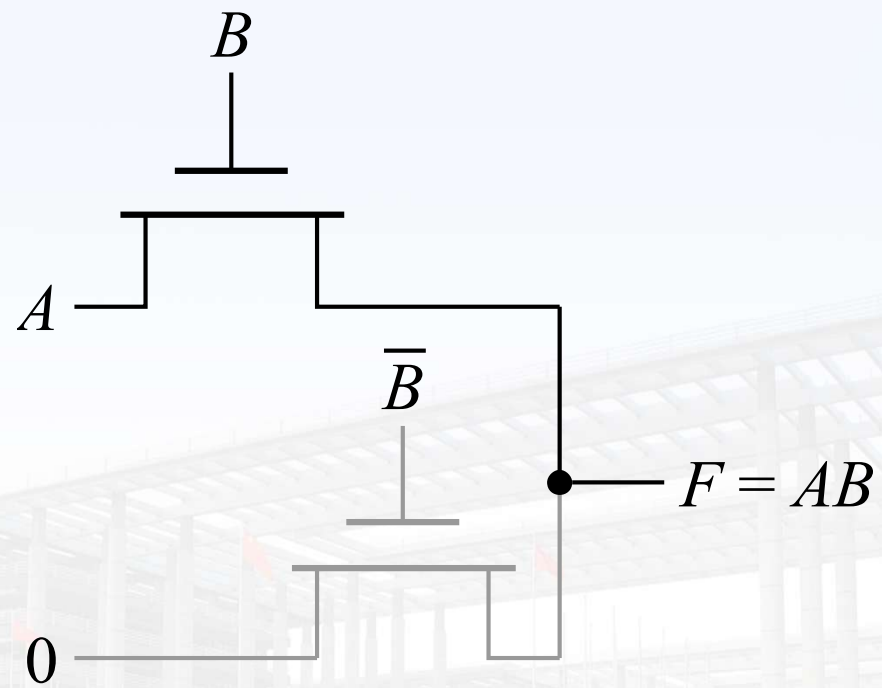


传输门逻辑

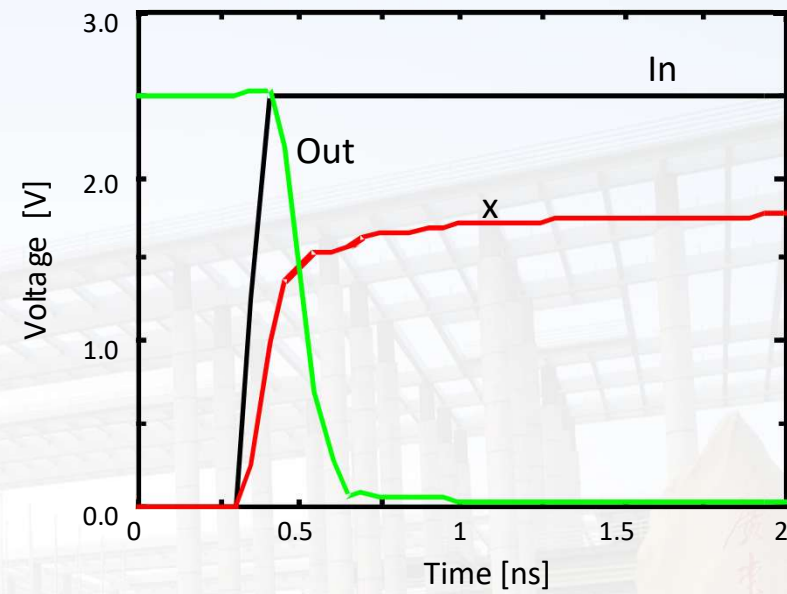
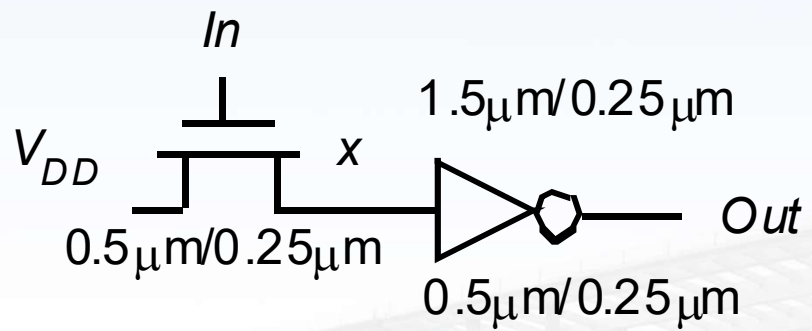


N个晶体管
无静态功耗

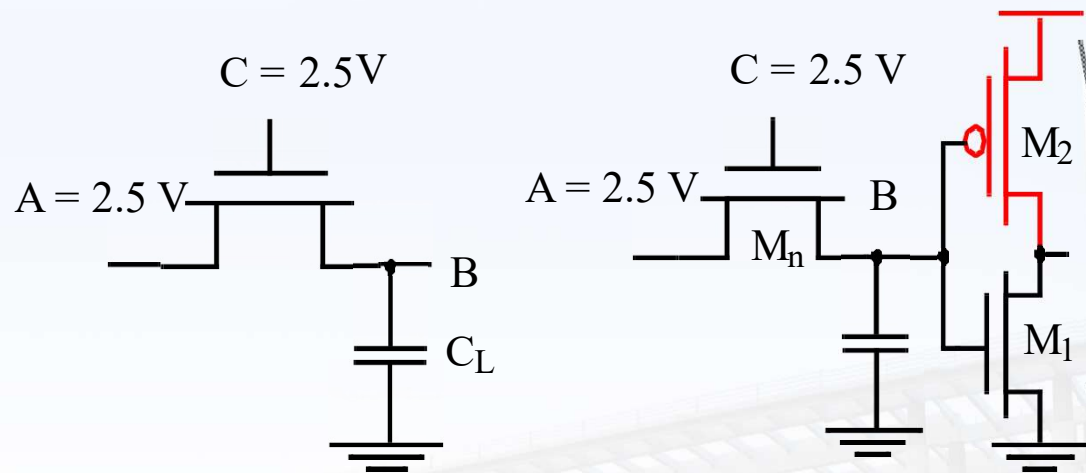
AND门



仅NMOS逻辑



仅NMOS开关电路



- 受阈值影响， V_B 无法提升到A点电压，只能到 $V_A - V_{TH}$
- 阈值电压的损失带来了静态功耗（无法M2管关断）
- NMOS的阈值电压比PMOS更高

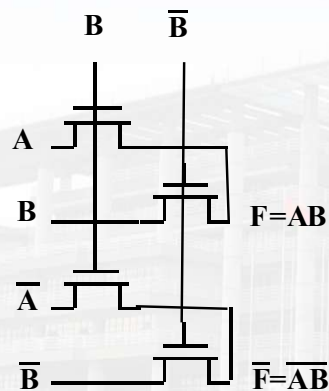
互补传输门逻辑



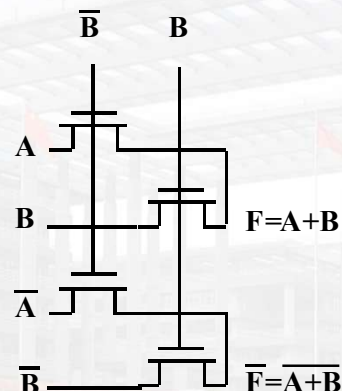
(a)



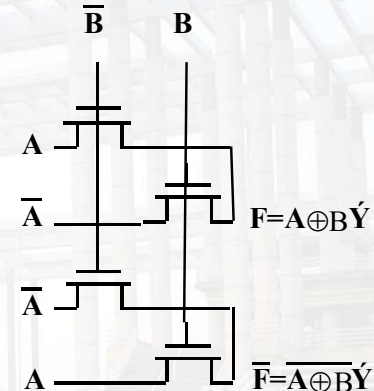
优势：全差分电路
静态门电路
模块化设计



AND/NAND



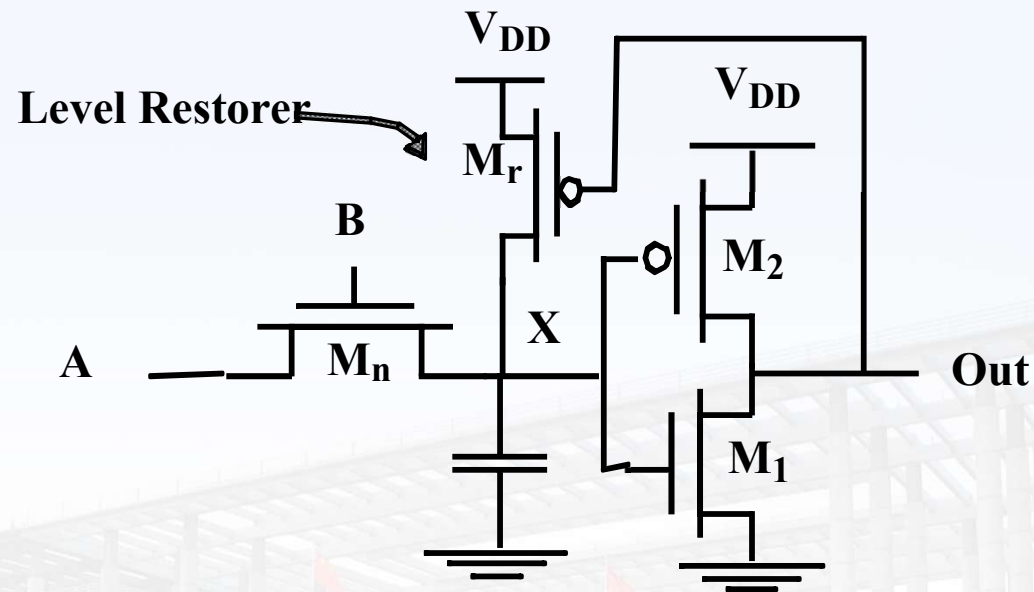
OR/NOR



EXOR/NEXOR

(b)

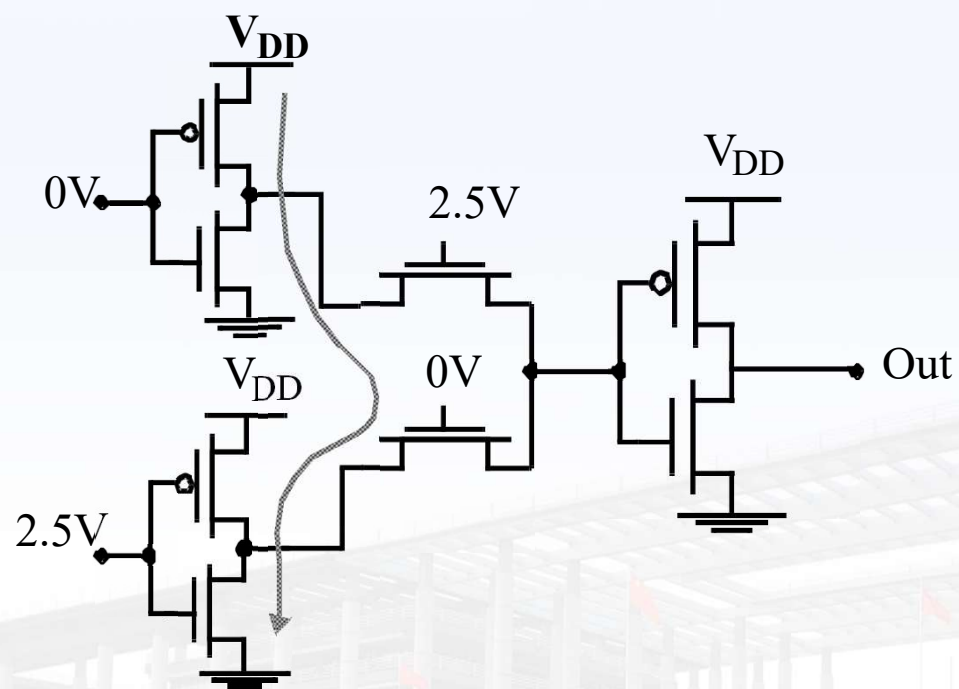
仅NMOS逻辑—电平恢复



优势：全摆幅

问题：1.X点电容增加，门上升时间受到影响
2.有比电路

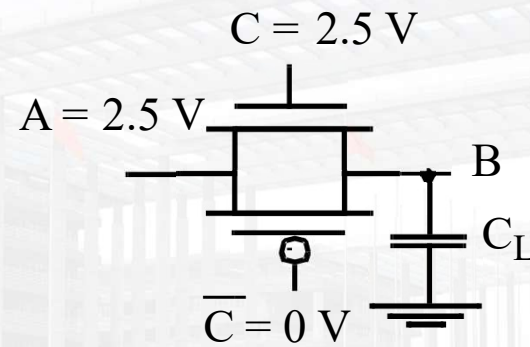
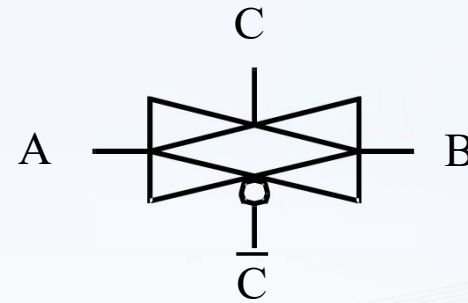
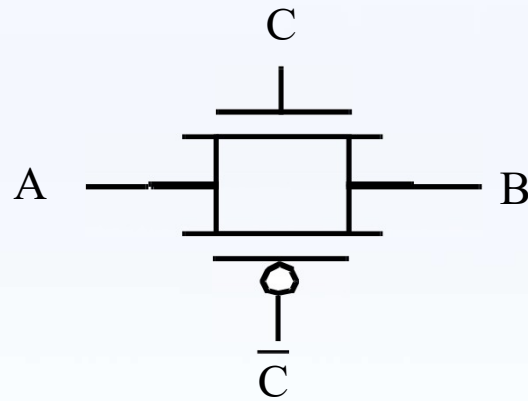
多种阈值晶体管



泄露电流

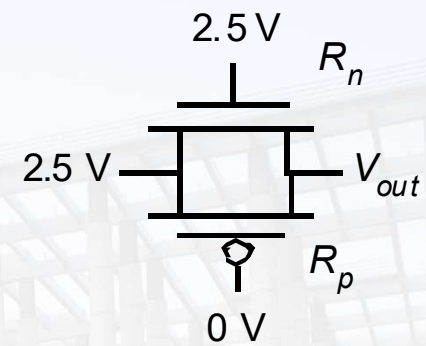
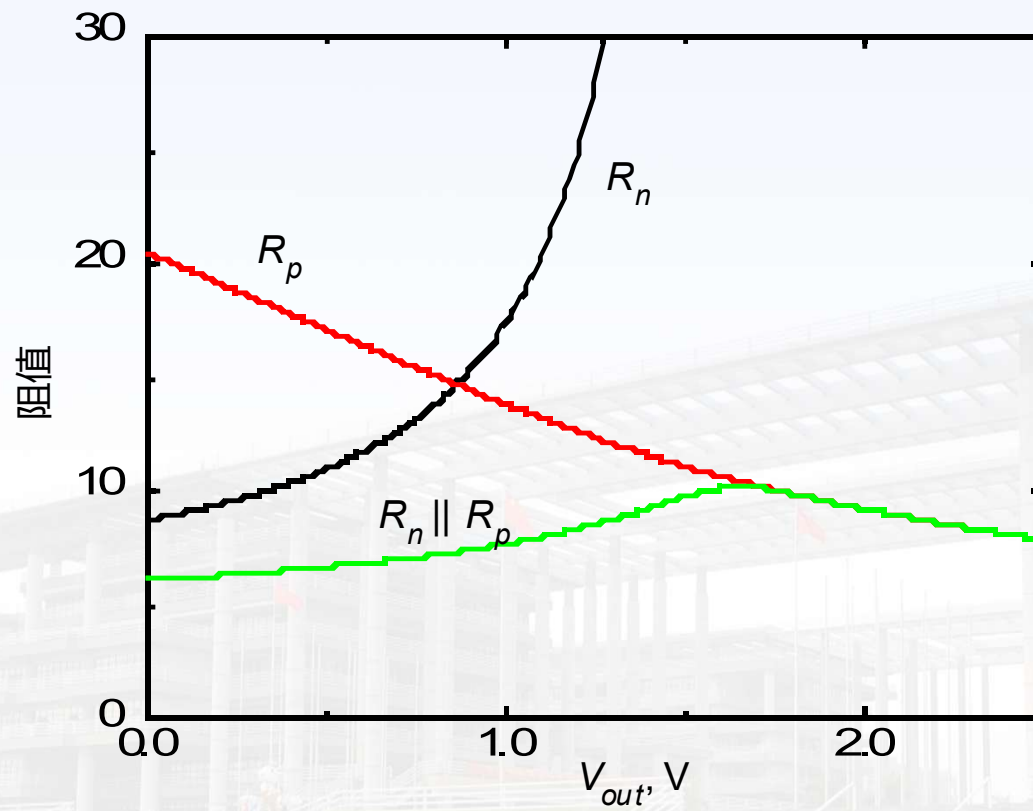
引入零阈值NMOS器件，降低阈值损失，但是会存在亚阈值电流导致的静态功耗。

传输门逻辑

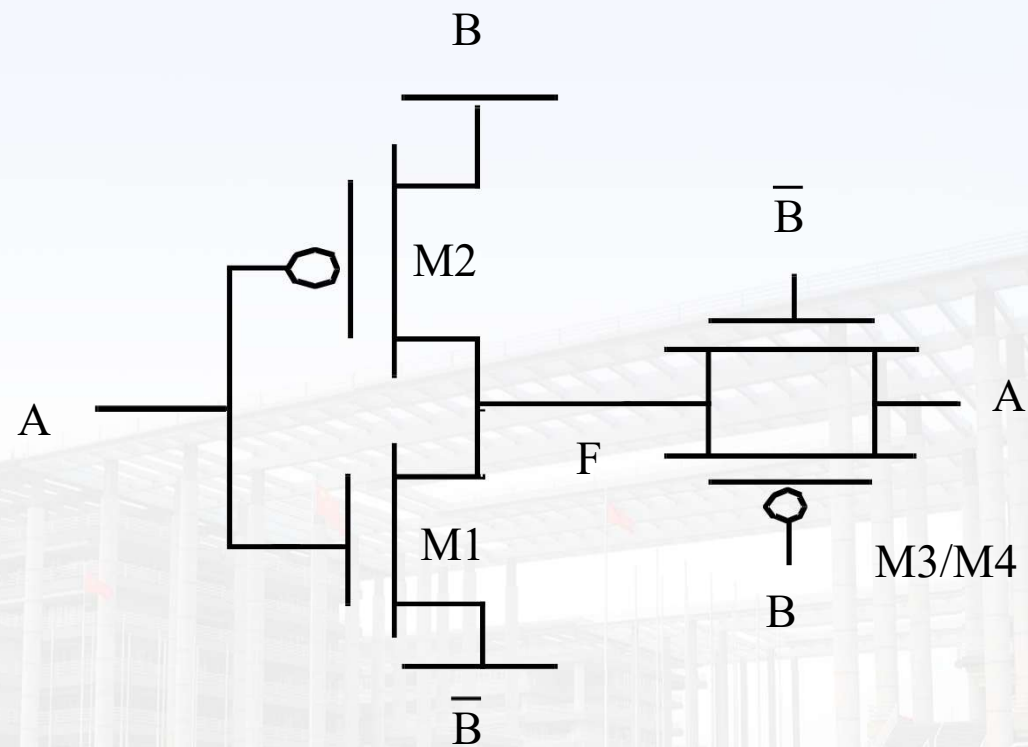


若 $C=1$ ，则 $A=B$

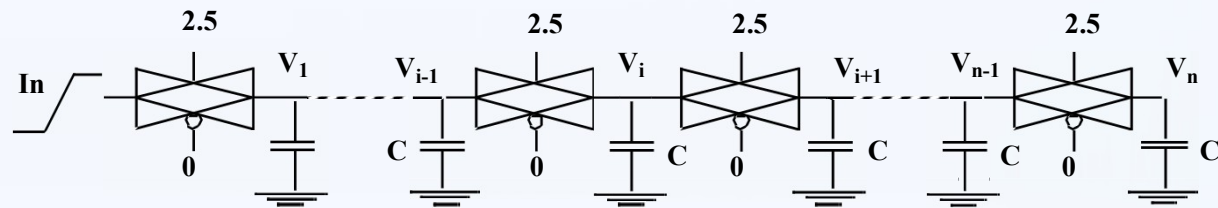
传输门的阻值



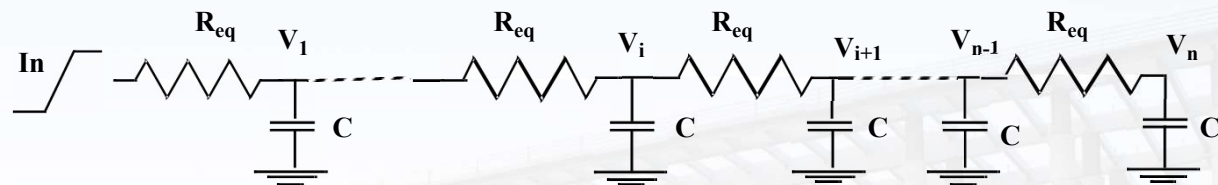
传输门XOR



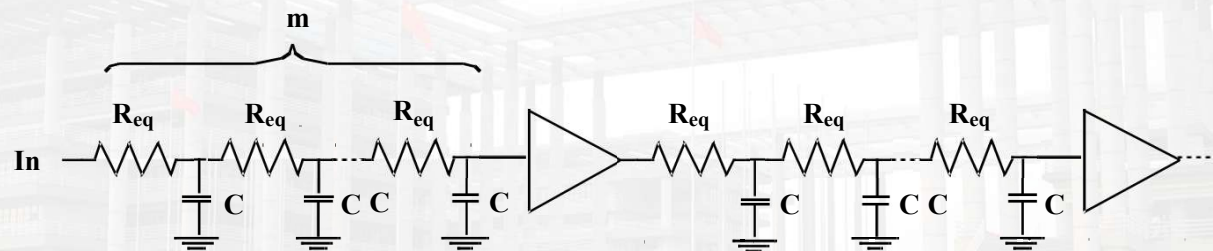
传输门网络中延时



(a)



(b)



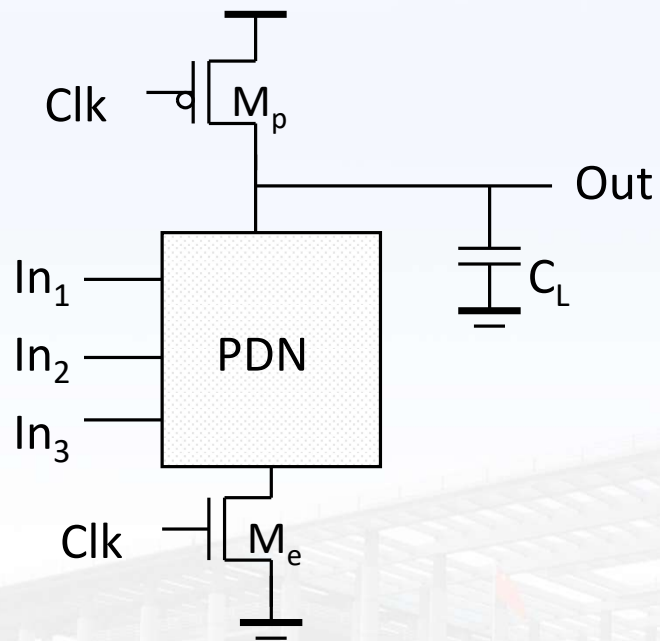
(c)



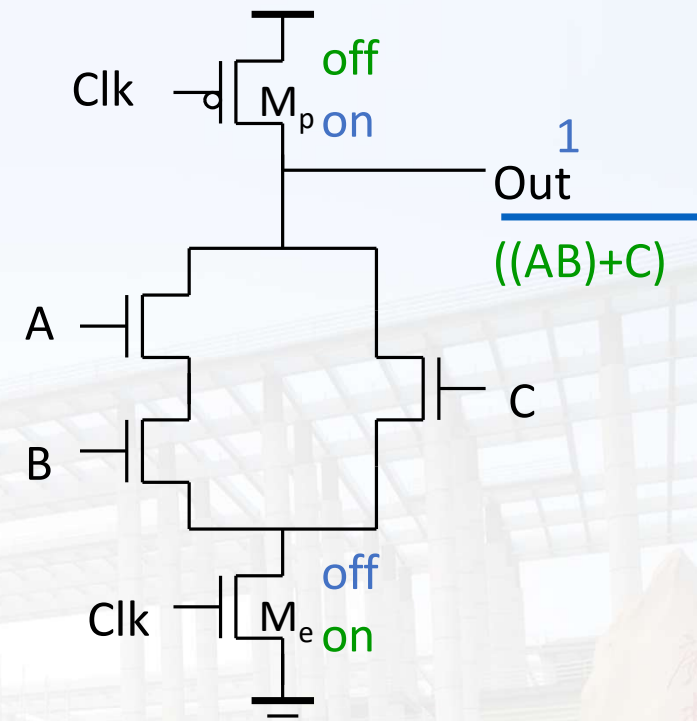
动态CMOS电路

- 在静态CMOS电路中，输出需要通过一个低阻的通道（低延时）连接到GND或 V_{DD} ：
 - 优势：电路结构简单
 - 不足：N输入电路需要 $2N$ 个晶体管
 - 改进：伪NMOS逻辑、传输门逻辑等
- 动态电路改变对低阻通道的依赖，采用高阻节点电容上存储信号值
 - 仅需要 $n+2$ ($n+1$ NMOS + 1 PMOS)晶体管

动态逻辑

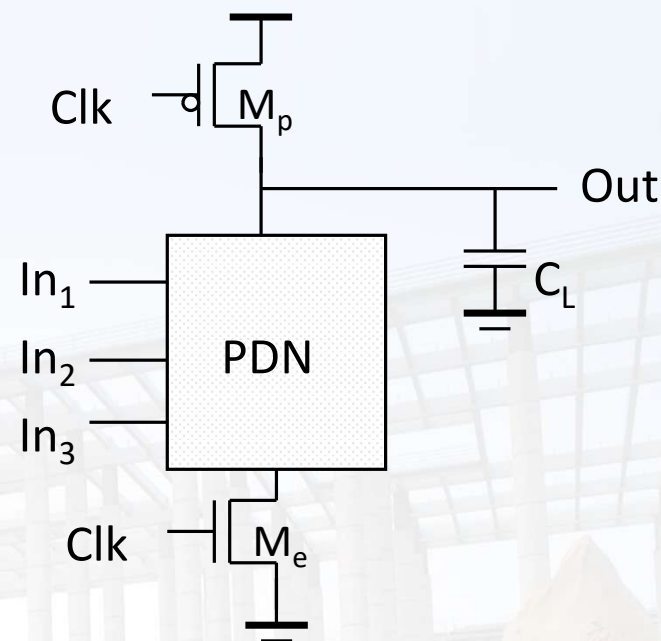


工作阶段：
预充电 ($CLK = 0$)
求值 ($CLK = 1$)



输出条件

- 当动态电路中输出电容被放电后，在下一个充电操作前是无法被再次充电的；
- 在求值阶段，门输入信号最多进行一次变化；
- 在求值期间和结束后，输出可以处于高阻态电路，状态被存储在负载电容 C_L 中。

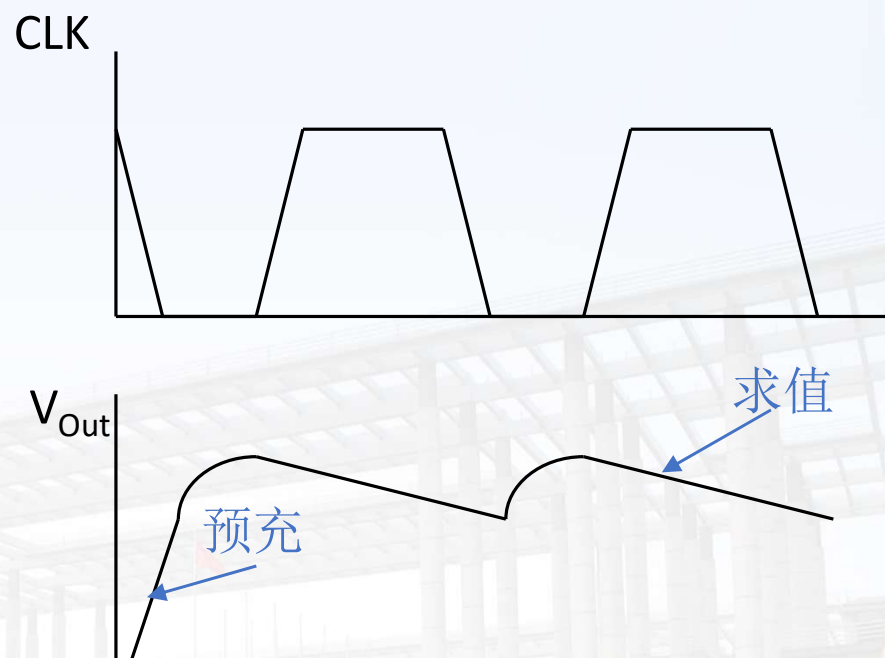
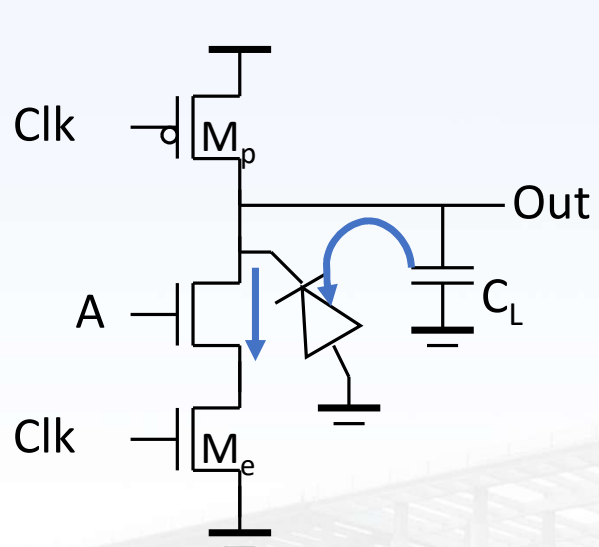




动态电路特性

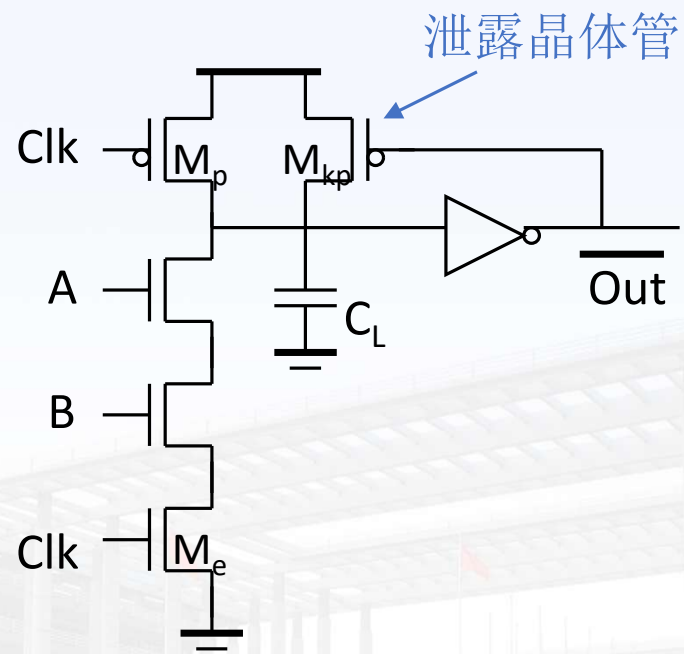
- 逻辑实现仅依赖NMOS
 - 晶体管数量减少到 $N + 2$ (对比: $2N$ 静态互补 CMOS)
- 全摆幅电路 ($V_{OL} = \text{GND}$ and $V_{OH} = V_{DD}$)
- 无比电路-器件尺寸变化不影响逻辑值
- 只有动态功耗, 但会高于静态CMOS
- 更快的转换速度
 - 减小了负载电容
 - 1. 更小的输入电容 (仅PDN)
 - 2. 更小的输出负载
 - 没有短路电流, 所有电流均流入到 C_L

电荷泄露



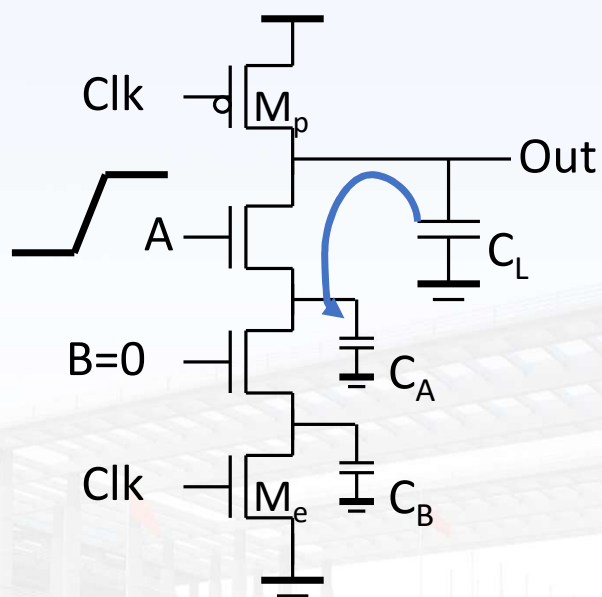
电荷泄露

电荷泄露解决方案



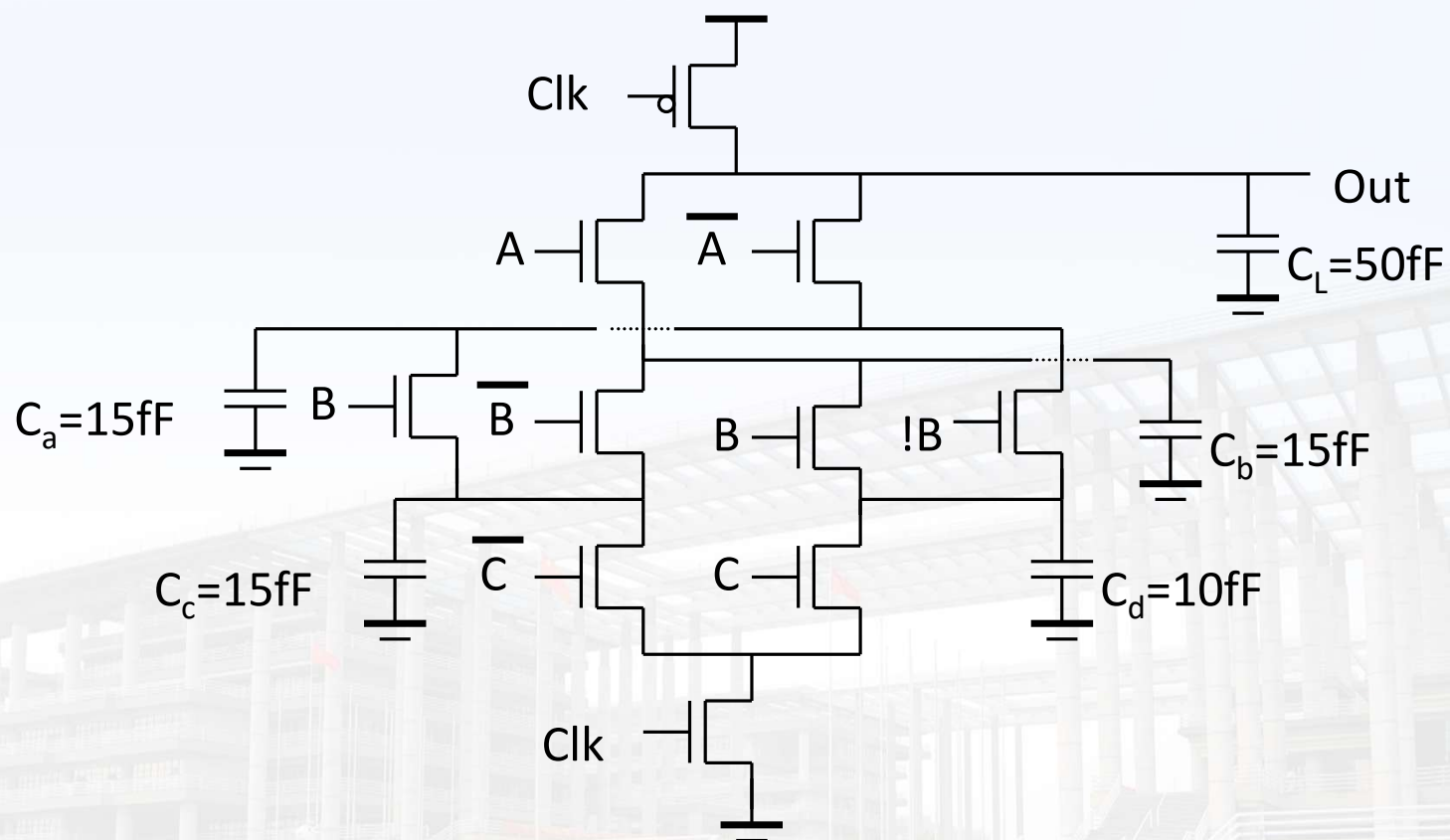
增加泄露晶体管，弥补下拉漏电路径造成的电荷损失。

电荷分享

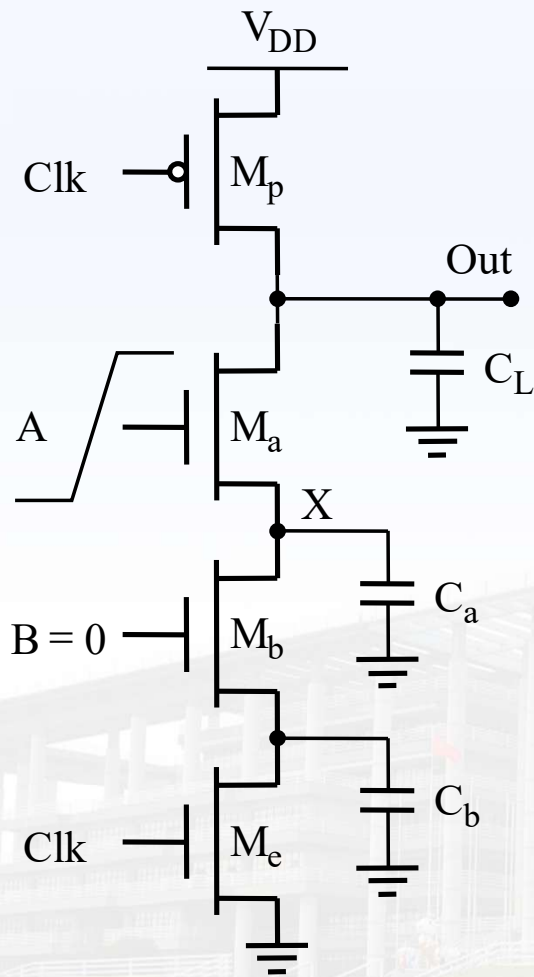


原本存储在 C_L 的电荷被 C_L 和 C_A 分享了，降低了输出电压

电荷分享



电荷分享



1) 当 ΔV_{out} 小于NMOS阈值

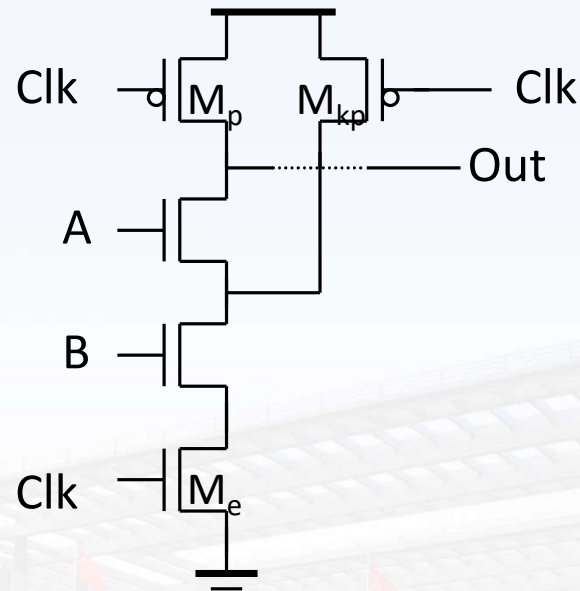
$$C_L V_{DD} = C_L V_{out}(t) + C_a (V_{DD} - V_{Tn}(V_X))$$

or

$$\Delta V_{out} = V_{out}(t) - V_{DD} = -\frac{C_a}{C_L} (V_{DD} - V_{Tn}(V_X))$$

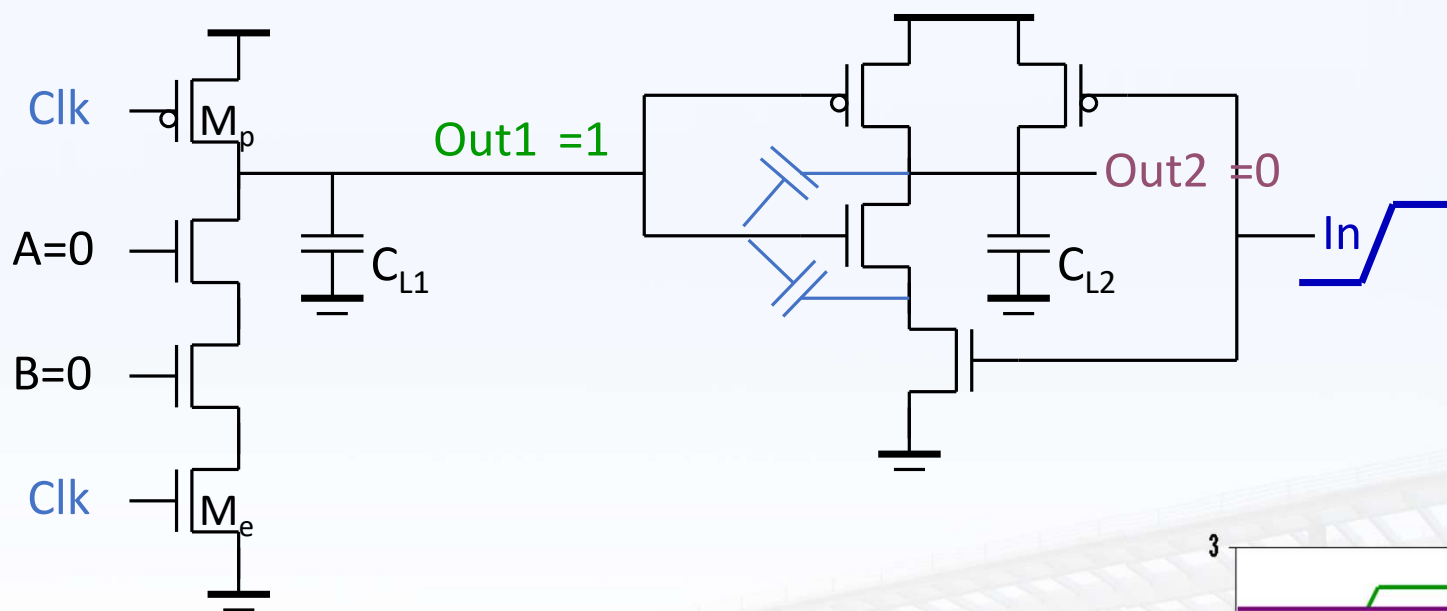
2) 当 ΔV_{out} 大于NMOS阈值

$$\Delta V_{out} = -V_{DD} \left(\frac{C_a}{C_a + C_L} \right)$$



使用时钟驱动的晶体管对内部节点进行预充电
问题：1.增加了面积和功率；2.进一步加大了时钟负载

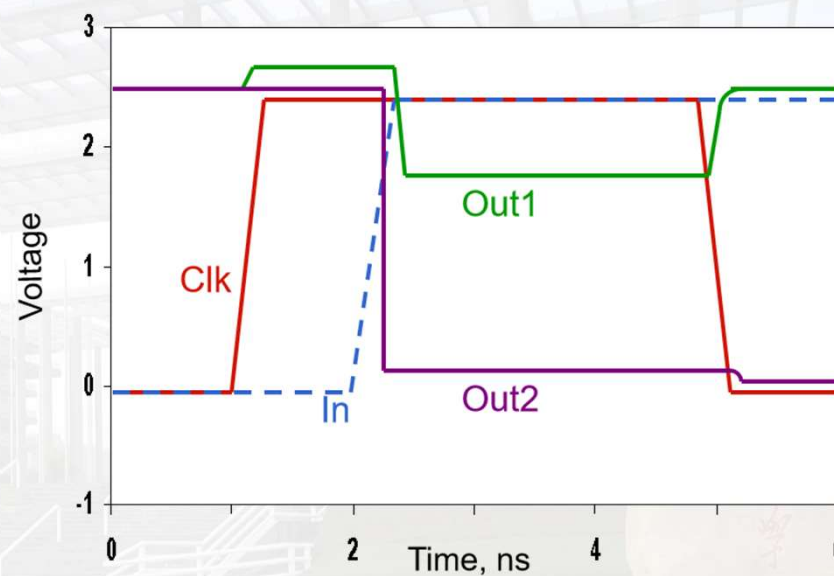
电容耦合



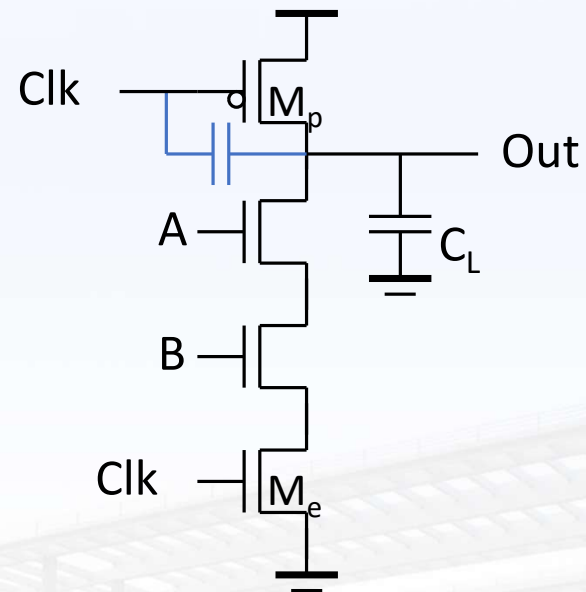
动态NAND

静态NAND

In信号的变化导致Out2节点拉低，通过电容耦合到Out1节点，导致NAND门无法彻底关闭

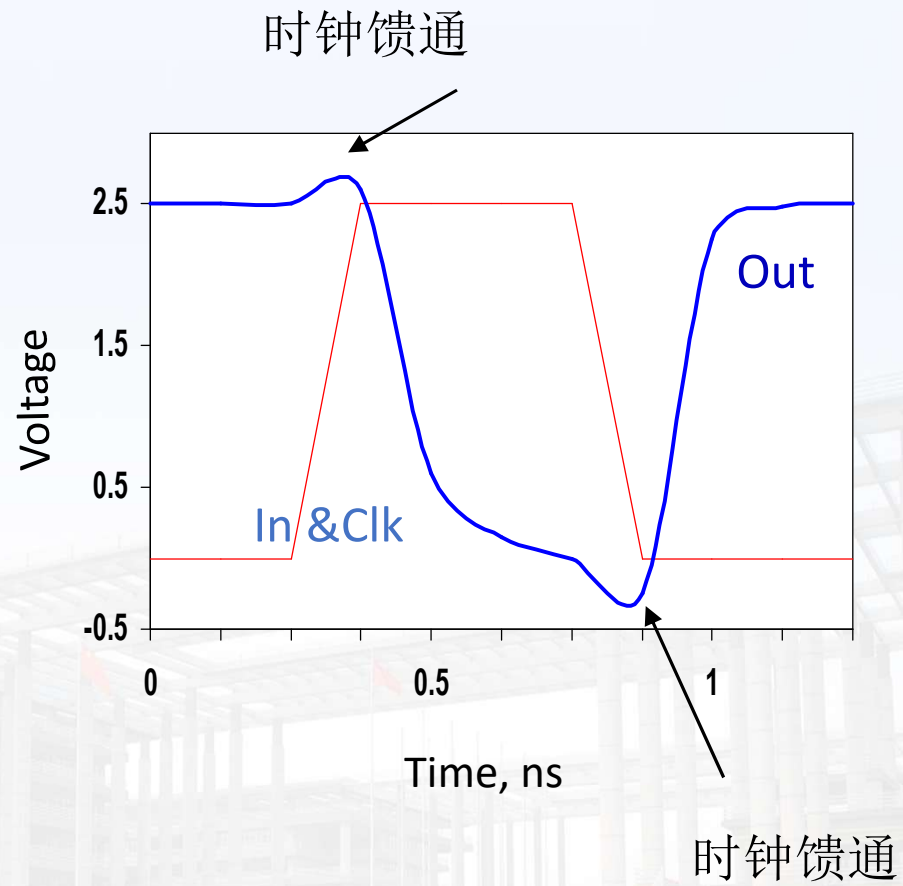
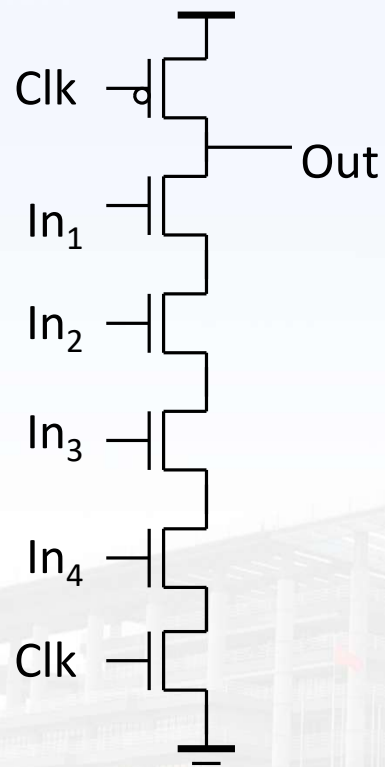


时钟馈通

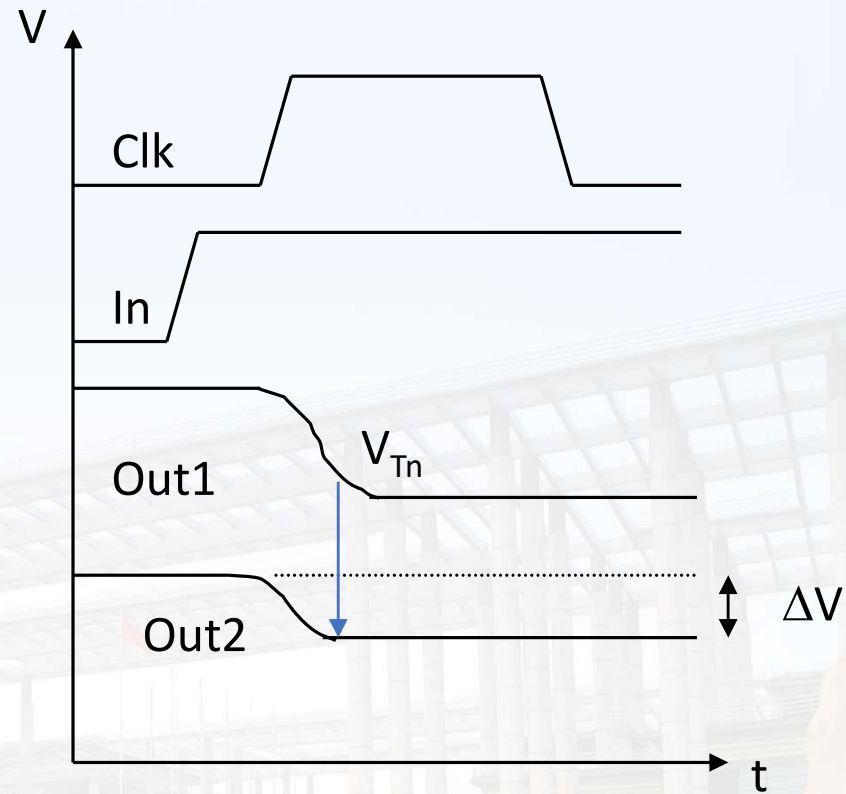
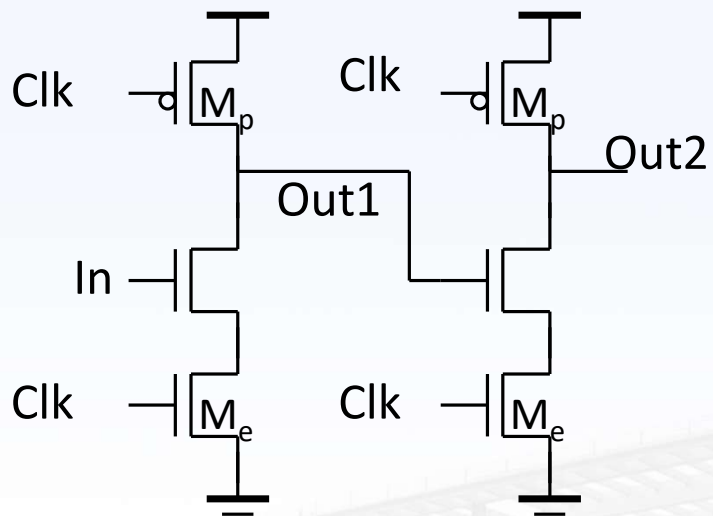


由于栅极到漏极的电容，预充电器件的 Out 和 Clk 输入之间存在耦合。所以 Out 的电压可以上升到 VDD 以上。时钟的快速上升沿（和下降沿）耦合到 Out 。

时钟馈通

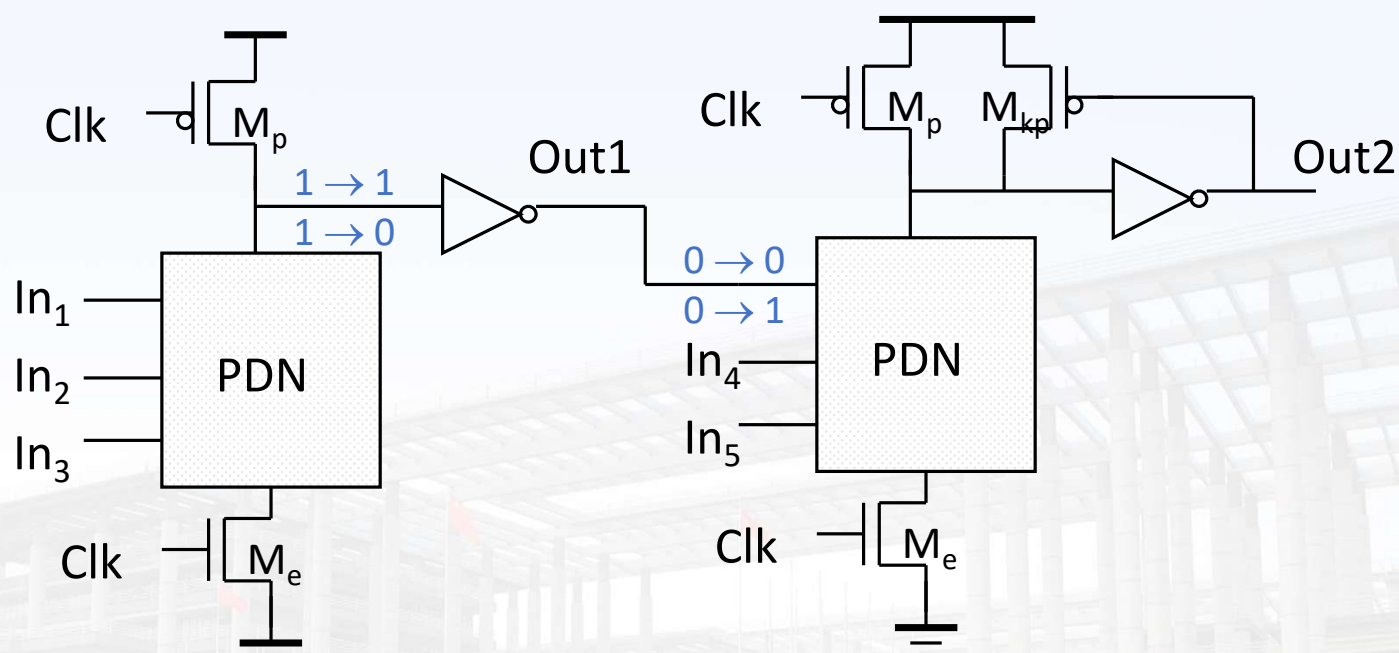


动态逻辑的级联

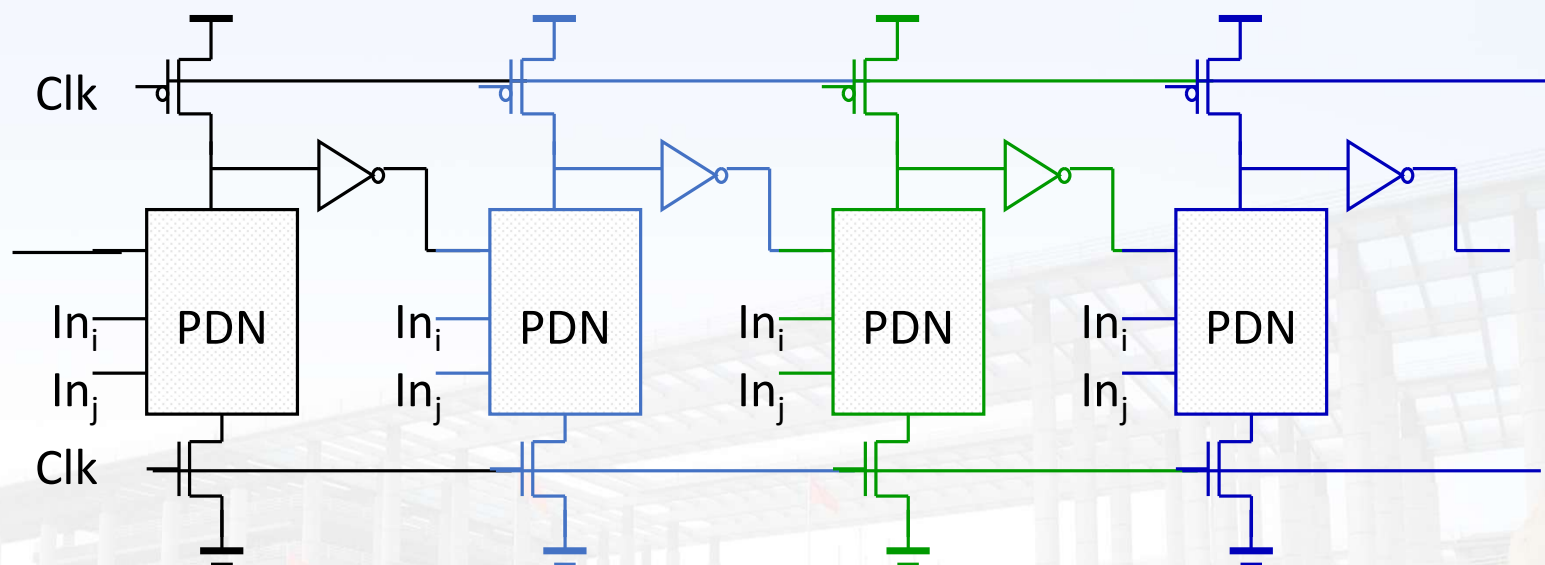


仅允许输入端存在 $0 \rightarrow 1$ 跳变

多米诺逻辑



多米诺逻辑

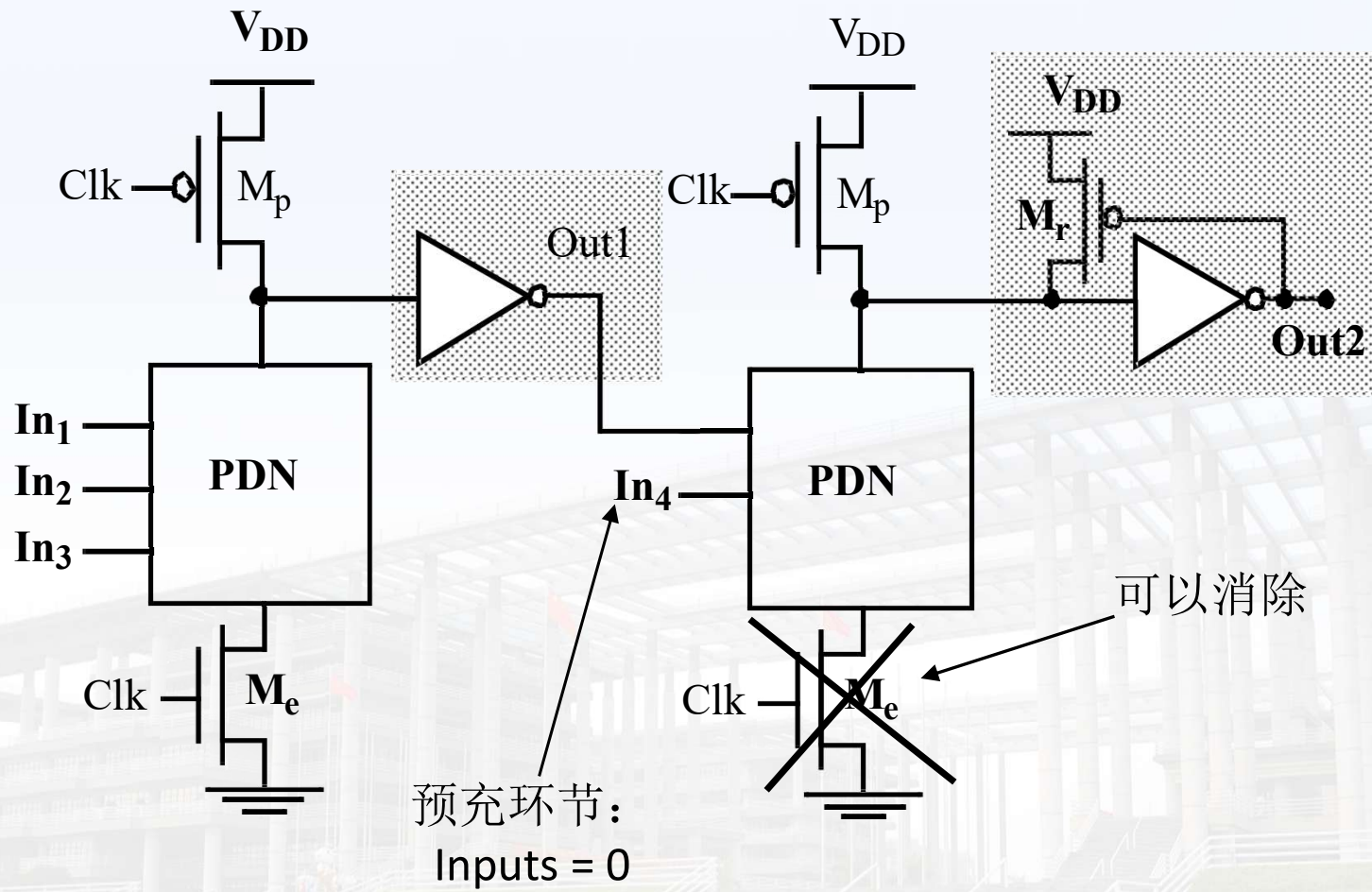




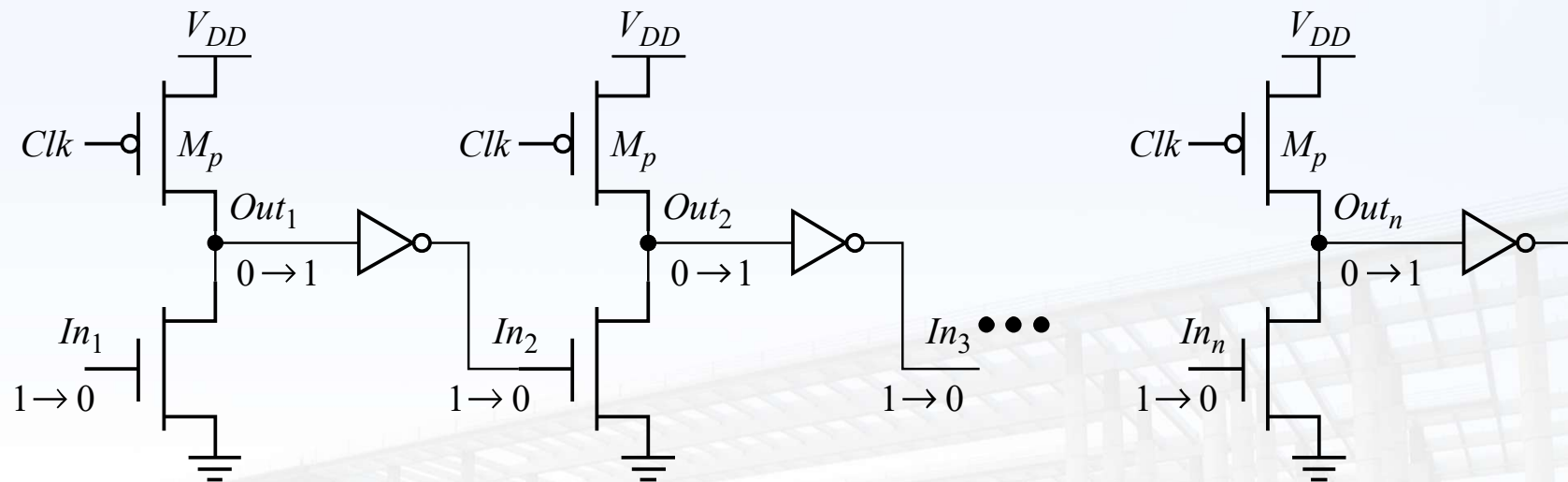
多米诺逻辑特性

- 由于每一级输出都插入了反相器，导致仅能够实现非反向逻辑
- 高速：
 - 仅有L-H转换，平均延时会更低
 - 反相器实现了隔离，减小了输入电容，扇出仅有一个输入栅电容

多米诺逻辑设计

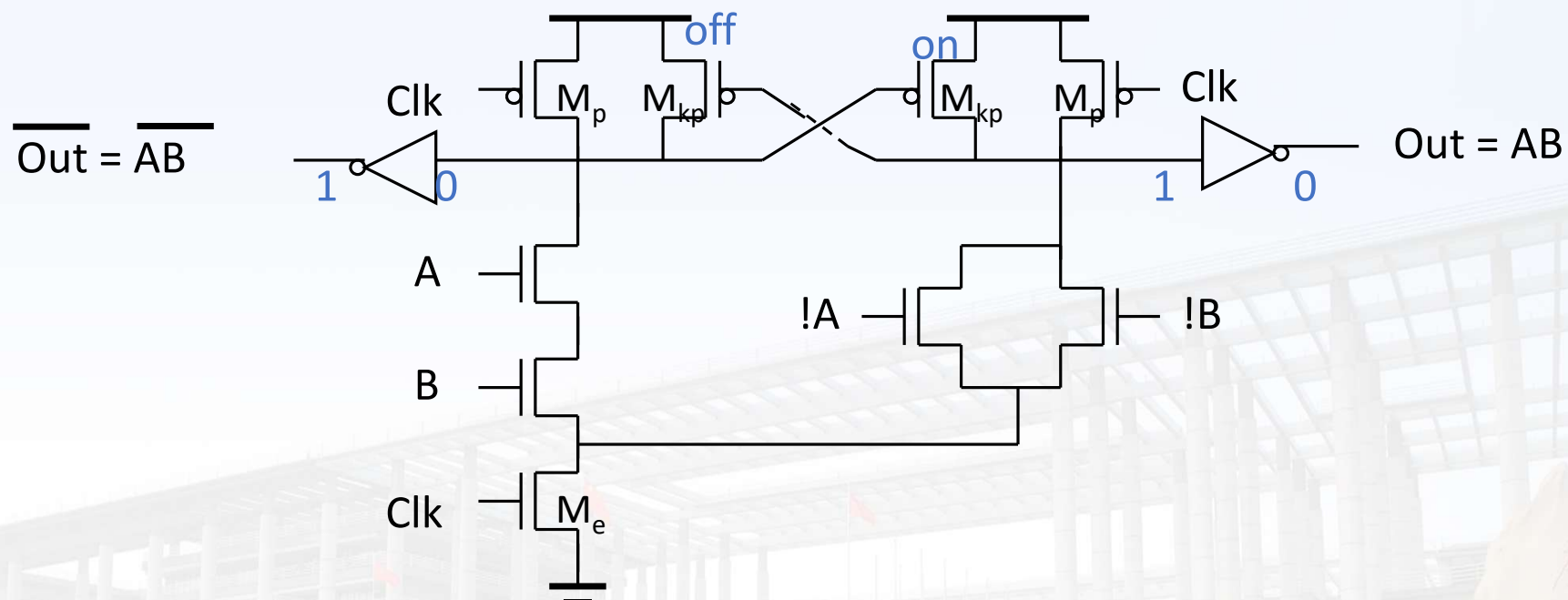


取消求值晶体管



延时较大，充电需要从第一级向后传递

差分多米诺结构

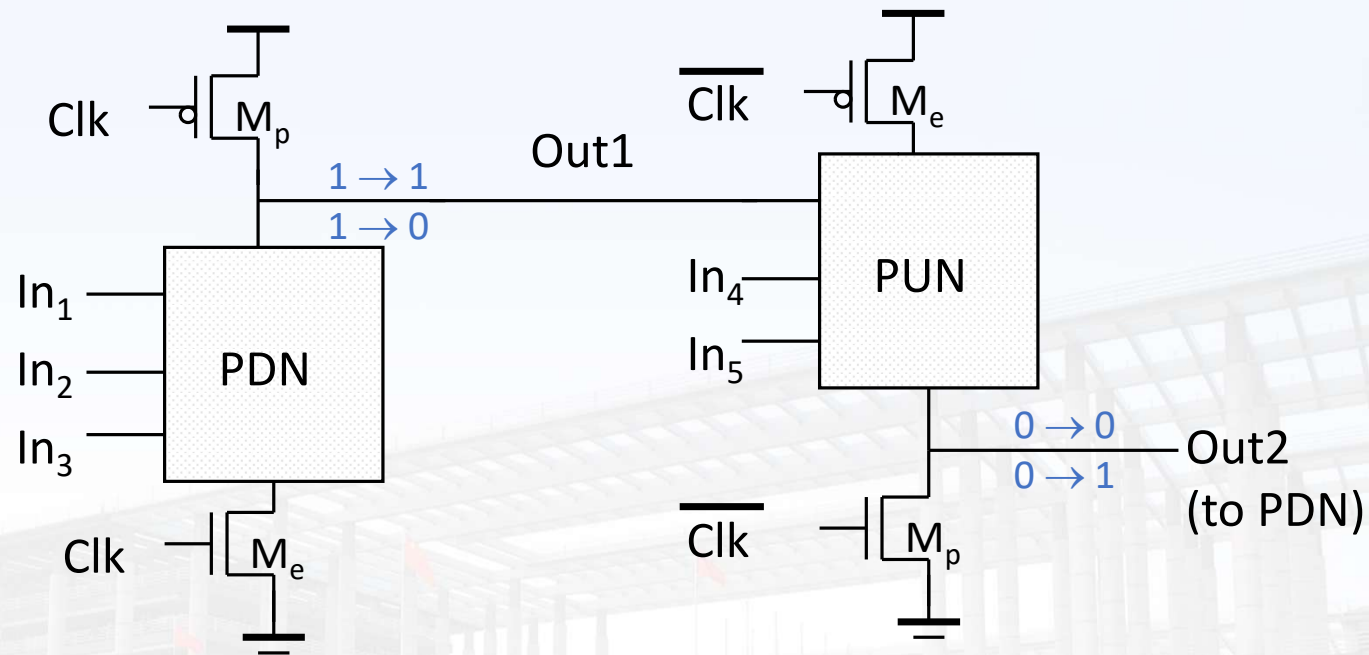


针对多米诺逻辑中非反向问题，引入差分结构。

优势：速度快

劣势：功耗大

np-CMOS



交替采用PDN和PUN避免反相器的插入

设计选择

- 设计目标：
 - 速度
 - 面积
 - 功耗
 - 稳定性
- 选型：
 - 静态电路
 - 互补CMOS电路
 - 伪NMOS电路
 - 传输门电路
 - 动态电路



谢谢！ 请批评指正！



广东工业大学集成电路学院

GUANGDONG UNIVERSITY OF TECHNOLOGY
SCHOOL OF INTEGRATED CIRCUITS